DQW PROJECT

Discrete-Time Quantum Walks in a Rectangular Billiard

From the shift and the geometry
to the coin and the full Floquet operator

Consolidated teaching notes

From established constructions to correctly qualified spectra

Revision: the 16-cell coin map and the routes out of exact solvability

The question guiding the notes

Which structures are fixed by the shift and the rectangle, which of them survive a chosen coin, and what can those surviving structures tell us about exact solvability and spectral statistics?

29 October 2026

A reading guide for the student

A coin, a transport rule, and a domain play different roles. These notes construct the complete quantum walk in stages. The first four stages identify what is present before a coin has been selected; the last three read every coin through the same geometric lens and then ask how each one breaks (or keeps) the exact solvability of the rectangle.

Stage	Question	Ingredients fixed so far
1. Floquet structure	What do the two factors do?	A finite position space and four internal states; no specified shift or coin
2. Shift	What happens on an allowed or blocked move?	A moving, stay-and-reverse prescription on a general site set
3. Rectangle	Which cycles and reflections follow?	The rectangle and the shift, with no coin chosen
4. The 16-cell map	What does the wall do to a pair of directions, and which coin amplitudes keep a standing wave intact?	Coherent channels $h_\pm, v_\pm$, their spatial envelopes, and the allowed/forbidden cells of a 4×4 coin
5. Local coins	What does each explicit matrix do, read in the 16-cell map?	Identity, permutation, Grover, Fourier, two Bell matrices, modified Grover, modified Bell
6. Full walks	Which structures survive, and what do the spectra and eigenstates show?	Their ordered composition with the same rectangle and walls
7. Mechanisms	By which route does each coin leave exact solvability?	The demonstrated reduction families and the coins placed among them

What is new in this revision. (i) Chapter 4 is new. It defines the coherent channels $h_\pm, v_\pm$ (the combinations of two opposite directions that a wall returns as themselves) and the envelope of a state (its spatial profile). It shows why the question "does a coin keep the standing waves of the rectangle?" reduces to the sixteen entries of the coin matrix. It also states which bases may be used to draw that 4×4 square: any basis gives a valid square, but only the coherent basis carries the allowed/forbidden reading. Every coin is then drawn in this form. (ii) The designed (modified) Fourier family is removed; the unperturbed Fourier coin is added, together with an explicit gauge transformation proving that its compass assignment is exactly solvable. (iii) Two Bell matrices are now distinguished: B_{I}, the matrix of the earlier report, and B_{II}, the matrix that reproduces the project's Bell IPR figure. The modified Bell family $B_{\mathrm{II}} e^{-i\theta V}$ is explained. (iv) The latest results computed for this revision (IPR reproductions, the 31×23 Grover and Bell sweeps, symmetry searches) are included with their figures, collected in the folder `Figures`. (v) Chapter 7 assembles the different routes to the breaking of integrability observed so far. Whether these routes are one mechanism in disguise, for the fixed geometry and shift, is stated as an open question and deliberately not resolved here.

Chapters 1–7 form the main teaching sequence. The appendices keep the wider established record: gauge equivalences, advanced diagnostics, supplementary tested constructions and the older random-coin interpolation.

We assume finite-dimensional linear algebra, complex amplitudes, tensor products and elementary quantum mechanics. Direct sums, projectors, coherent coin combinations and antiunitary operations are explained where needed.

One main convention. Directions are ordered (R, L, U, D). The same 4×4 coin acts first at every site; the shift acts second. On an existing link it keeps the direction label. On a blocked move it stays at the site and reverses direction with amplitude $+1$. No periodic boundary or extra reflection phase is used. The older flip-flop/self-loop model is kept separate (Appendix D).

Evidence labels. An *exact result* is an algebraic identity or construction displayed in the text. A *recorded numerical result* was reported as executed in an earlier project export, with its geometry, resolution and spectral selection. Results marked *computed for this revision* were obtained with an independent Python/NumPy implementation of exactly the convention above; their sizes and filters are stated where they are used. A verified value of a statistic does not automatically verify a broader interpretation. The complete classification of either Bell walk remains open.

Scope. The core domain is a nonsquare rectangle; square controls are labeled, disk and triangle results are outside scope. Appendix E gives the source and figure inventory.

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Chapter 1

The Floquet operator before choosing its ingredients

1.1 Position and the four internal states

Let Ω be a finite set of N allowed positions. At this stage its shape has not been specified. A basis vector $|\mathbf{r}, d\rangle$ states both where the walker is and which internal state it carries. The four internal labels will later be associated with directions, in the fixed order

$$(R, L, U, D).$$

The state space is

$$\mathcal{H} = \mathcal{H}_{\text{pos}} \otimes \mathcal{H}_{\text{coin}}, \qquad \mathcal{H}_{\text{coin}} = \mathbb{C}^4, \qquad \dim \mathcal{H} = 4N \equiv D. \tag{1.1}$$

The tensor product means that a basis state requires *both* a position label and an internal label. A general normalized state is

$$|\psi_n\rangle = \sum_{\mathbf{r}\in\Omega} \sum_{d} \psi_d(\mathbf{r}, n)|\mathbf{r}, d\rangle, \qquad \sum_{\mathbf{r},d} |\psi_d(\mathbf{r}, n)|^2 = 1. \tag{1.2}$$

These coefficients are complex amplitudes. Different contributions to the same final basis state add before their absolute square is taken. This is why probabilities alone will not determine the walk.

1.2 The onsite coin and the shift

The coin is a unitary 4×4 matrix C. Throughout the main model it is homogeneous: the same matrix acts at every site and every step. Its full-space action is

$$\mathcal{C} = I_{\text{pos}} \otimes C, \qquad \mathcal{C}|\mathbf{r}, a\rangle = \sum_{d} C_{da}|\mathbf{r}, d\rangle. \tag{1.3}$$

Rows of C label outputs and columns label inputs. The coin changes the internal amplitudes at a fixed position.

The shift S is a second unitary operator on $\mathcal{H}$. Its matrix has dimension $D \times D$, not 4×4. We have not yet specified what move it implements or how it treats a boundary. One complete step is

$$\boxed{U = S\mathcal{C} = S(I_{\text{pos}} \otimes C), \qquad |\psi_{n+1}\rangle = U|\psi_n\rangle.} \tag{1.4}$$

Operators act from right to left: coin first, shift second. For any input,

$$U|\mathbf{r}, a\rangle = \sum_d C_{da} S|\mathbf{r}, d\rangle. \tag{1.5}$$

The coin supplies local amplitudes and the shift transports their basis states. This separates their jobs before a move is specified. A coin is not a measurement and need not be random. In general $S\mathcal{C} \neq \mathcal{C}S$, so the order must be stated.

1.3 Why the repeated step is called Floquet evolution

The same update is repeated at each integer time, so after n steps $|\psi_n\rangle = U^n|\psi_0\rangle$. We call U the Floquet, or one-period, operator. Its eigenvalues are phases:

$$U|\phi_j\rangle = e^{i\varphi_j}|\phi_j\rangle, \qquad \varphi_j \equiv \varphi_j + 2\pi. \tag{1.6}$$

If a period T and $\hbar = 1$ are introduced, the convention $U = e^{-iH_{\text{eff}}T}$ identifies $\varepsilon_j = -\varphi_j/T$, modulo $2\pi/T$. The walk is specified directly by its step; no particular continuous Hamiltonian or continuous-time approximation is assumed.

1.4 What follows from unitarity alone

Because both factors are unitary,

$$U^\dagger U = \mathcal{C}^\dagger S^\dagger S\mathcal{C} = I. \tag{1.7}$$

Thus probability is conserved, evolution is reversible, and U has an orthonormal eigenbasis. Its eigenphases lie on a circle. These are the general conclusions available before specifying either factor. Unitarity alone gives neither spatial separability nor a particular reflection symmetry, time-reversal symmetry, or spectral statistics.

1.5 Conserved subspaces and direct sums

A conserved observable Q satisfies

$$[U, Q] = UQ - QU = 0. \tag{1.8}$$

Then its expectation value is constant. A conserved subspace can be described by an orthogonal projector P:

$$P^2 = P = P^\dagger, \qquad [U, P] = 0. \tag{1.9}$$

For a unitary U, this means that both the subspace and its orthogonal complement are invariant.

One sufficient way to conserve Q is to have $[S, Q] = [\mathcal{C}, Q] = 0$. This is not a necessary condition for an arbitrary Q: the two factors can also combine to preserve a structure neither preserves separately. We must examine the full step.

For mutually orthogonal conserved projectors summing to identity,

$$\mathcal{H} = \bigoplus_a \mathcal{H}_a, \qquad \mathcal{H}_a = \operatorname{im} P_a, \qquad U = \bigoplus_a U_a.$$

The image is the subspace selected by the projector. This groups alternative state components, not different particles. A tensor product still specifies the labels of a state.

Every finite unitary has its eigenprojectors. Their existence alone cannot distinguish exact solvability from complicated dynamics. The project's useful reductions are constructed from transport, geometry and the coin before diagonalizing the full walk. A few parity blocks whose dimensions grow with the area are not a complete solution.

1.6 Keeping the operator sizes distinct

Object	Space on which it acts	Matrix size
Local coin C	Four internal states	4×4
Local direction observable q or reversal J	Four internal states	4×4
Full coin $\mathcal{C} = I_{\mathrm{pos}} \otimes C$	Position and direction	$D \times D$
Shift S, full step U, full symmetry or projector	Position and direction	$D \times D$
Restricted operator U_a	A sector of rank r_a	$r_a \times r_a$

The rank of a projector is the dimension of its image, not the number of rows of the projector matrix. A local direction operation is extended by the position identity before it is compared with the shift. These distinctions will be important when a four-dimensional coin block is compared with a full-space standing-wave block.

What we have learned

A walk state carries position and coin information. The coin acts onsite, the shift implements transport, and their ordered product is the Floquet step. Exact solvability or quantum-chaotic spectral behavior is a property of that product with its boundary convention.

Chapter 2

The shift: transport, boundaries, and reversal

2.1 One move and one boundary rule

We now choose the transport prescription while keeping the allowed-site set arbitrary. The notation S_Ω makes its domain dependence explicit: different site sets can give different cycles and symmetries.

Associate the internal labels with displacements

$$\delta_R = (1,0), \quad \delta_L = (-1,0), \quad \delta_U = (0,1), \quad \delta_D = (0,-1).$$

Let $\bar{d}$ be the opposite direction: $\bar{R} = L$, $\bar{L} = R$, $\bar{U} = D$, $\bar{D} = U$. The main shift is

$$\boxed{S_\Omega|\mathbf{r}, d\rangle = \begin{cases} |\mathbf{r} + \delta_d, d\rangle, & \mathbf{r} + \delta_d \in \Omega, \\ |\mathbf{r}, \bar{d}\rangle, & \mathbf{r} + \delta_d \notin \Omega. \end{cases}} \tag{2.1}$$

At an allowed move, the walker travels one site and keeps its direction. At a blocked move it stays where it is and reverses its direction. A boundary event therefore consumes one step of the shift. Every displayed amplitude is $+1$; no reflection phase or loss is included. The rule is applied linearly to each component of a superposition. Arriving at the last allowed site does not reverse the label. The reversal occurs on the attempted outward shift. There is no state at a nonexistent site.

The discrete geometry is the allowed-site set together with its nearest-neighbor links. A wall is represented by a missing link. There is no periodic identification. The rule does not require a continuous normal vector; in particular, a corner presents two different blocked directional channels.

2.2 Why this shift is unitary

The shift sends each position–direction basis state to precisely one basis state with amplitude one. It is also one-to-one. To find the predecessor of $|\mathbf{r}, d\rangle$, ask whether $\mathbf{r} - \delta_d$ is allowed. If it is, the predecessor is $|\mathbf{r} - \delta_d, d\rangle$. If it is not, the predecessor is $|\mathbf{r}, \bar{d}\rangle$, whose attempted move is blocked.

Thus S_Ω permutes the basis. Its matrix is real, with one entry one in each row and column:

$$\boxed{S_\Omega^{-1} = S_\Omega^{\mathsf{T}} = S_\Omega^\dagger.} \tag{2.2}$$

A finite permutation decomposes into disjoint cycles. Following a cycle here means repeatedly applying S_Ω *alone*. The coin will be inserted between successive shifts in the actual walk, and may transfer amplitude between these cycles.

2.3 What is fixed before choosing the shape?

The prescription already gives a real unitary permutation and finite basis cycles. It preserves the horizontal/vertical axis because opposite-direction reversal does not change the axis. Cycle lengths and spatial reflections depend on the site set. One further identity is independent of its shape.

2.4 Direction reversal reverses the shift

Define the local reversal matrix

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad J^2 = I_4. \tag{2.3}$$

Write $\mathcal{J} = I_{\text{pos}} \otimes J$. The established identity for the reflecting shift is

$$\boxed{\mathcal{J} S_\Omega \mathcal{J} = S_\Omega^{-1}.} \tag{2.4}$$

Reverse the direction, apply the move, and reverse again: this retraces the original move, including a blocked move. Importantly, Eq. (2.4) is a *reversing relation*, not a statement that $\mathcal{J}$ commutes with the shift.

Since the shift is real, complex conjugation K obeys $KSK = S$. Combining it with reversal gives $(\mathcal{J}K)S(\mathcal{J}K)^{-1} = S^{-1}$, with $(\mathcal{J}K)^2 = I$. This is a time-reversal relation for the shift alone. Chapter 6 explains why the corresponding test for the full walk must also include the coin.

The earlier sessions also used a different, involutive flip-flop shift. Its full comparison with this transport rule is retained in Appendix D. We keep the present prescription fixed throughout the main teaching sequence.

Chapter 3

The reflecting rectangle before a coin is chosen

3.1 Sites, dimensions, and the four edge rules

We choose the site set, not yet the coin. Unless a square control is identified, $N_x \neq N_y$. The indices count sites, rather than separately specified continuum side lengths. Fix

$$\Omega = \{1, \ldots, N_x\} \times \{1, \ldots, N_y\}, \qquad N = N_x N_y, \qquad D = 4N_x N_y. \tag{3.1}$$

For example, the horizontal rules become

$$S|x, y, R\rangle = \begin{cases} |x+1, y, R\rangle, & x < N_x, \\ |N_x, y, L\rangle, & x = N_x, \end{cases} \tag{3.2}$$

$$S|x, y, L\rangle = \begin{cases} |x-1, y, L\rangle, & x > 1, \\ |1, y, R\rangle, & x = 1. \end{cases} \tag{3.3}$$

The vertical rules are

$$S|x, y, U\rangle = \begin{cases} |x, y+1, U\rangle, & y < N_y, \\ |x, N_y, D\rangle, & y = N_y, \end{cases} \tag{3.4}$$

$$S|x, y, D\rangle = \begin{cases} |x, y-1, D\rangle, & y > 1, \\ |x, 1, U\rangle, & y = 1. \end{cases} \tag{3.5}$$

At a corner its two outward components reverse independently. There is no diagonal move. Horizontal motion preserves y; vertical motion preserves x, yielding the first invariant decomposition.

3.2 Rows and columns are independent cycles of the shift

For a fixed row and a fixed column define

$$\mathcal{H}_y^H = \mathrm{span}\{|x, y, R\rangle, |x, y, L\rangle : 1 \leq x \leq N_x\}, \tag{3.6}$$

$$\mathcal{H}_x^V = \mathrm{span}\{|x, y, U\rangle, |x, y, D\rangle : 1 \leq y \leq N_y\}. \tag{3.7}$$

Their dimensions are $2N_x$ and $2N_y$, respectively. They are mutually orthogonal because their position–direction basis states are distinct. They exhaust the full basis, so

$$\boxed{\mathcal{H} = \left(\bigoplus_{y=1}^{N_y} \mathcal{H}_y^H\right) \oplus \left(\bigoplus_{x=1}^{N_x} \mathcal{H}_x^V\right).} \tag{3.8}$$

Each of these spaces is invariant under S.

The row projectors are explicit full-space matrices:

$$P_y^H = \sum_{x=1}^{N_x} \sum_{d\in\{R,L\}} |x, y, d\rangle\langle x, y, d|, \tag{3.9}$$

and P_x^V is defined analogously. They satisfy $[S, P_y^H] = [S, P_x^V] = 0$ and sum to I_D. The coarser axis projectors are

$$P_H = \sum_y P_y^H = I_{\text{pos}} \otimes \text{diag}(1, 1, 0, 0), \qquad P_V = I - P_H.$$

On one row the basis states occur in the cycle

$$|1, y, R\rangle \to \cdots \to |N_x, y, R\rangle \to |N_x, y, L\rangle \to \cdots \to |1, y, L\rangle \to |1, y, R\rangle. \tag{3.10}$$

Its length is $2N_x$. Each column similarly has length $2N_y$.

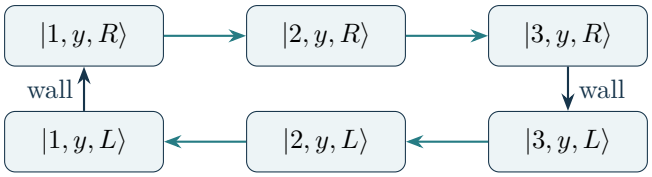


Figure 3.1: One shift cycle for a row of three sites. A wall arrow reverses the direction at the same site. The diagram describes repeated shifts; it does not yet include the action of an intervening coin.

If a cycle has basis $|j\rangle$, $j = 0, \ldots, L-1$, with $S|j\rangle = |j + 1 \bmod L\rangle$, its Fourier vectors are

$$|k\rangle = \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} e^{-2\pi ikj/L}|j\rangle, \qquad S|k\rangle = e^{2\pi ik/L}|k\rangle. \tag{3.11}$$

Thus the rectangular shift is exactly diagonalizable. Its order is

$$\boxed{S^{2\,\text{lcm}(N_x,N_y)} = I.} \tag{3.12}$$

The displayed exponent is its minimal positive period. It is not a periodicity statement for a general $U = S\mathcal{C}$.

3.3 Why a direct sum does not replace the tensor product

The same space can be written as

$$\mathcal{H} = \mathcal{H}_x \otimes \mathcal{H}_y \otimes \mathbb{C}^4.$$

This describes which labels specify a state. Equation (3.8) instead groups alternative sets of basis states on which the shift acts independently. These are compatible descriptions.

For a decomposition $|\psi\rangle = \sum_a |\psi_a\rangle$, $|\psi_a\rangle \in \mathcal{H}_a$, a block-diagonal operator evolves each component within its own $\mathcal{H}_a$. It does not describe several independent walkers. The dimension check is

$$N_y(2N_x) + N_x(2N_y) = 4N_xN_y.$$

Multiplying row-space and column-space dimensions would describe a different system.

3.4 Reflections must act on positions and directions

A geometric reflection must also reflect the direction vector. Define

$$\mathcal{P}_x|x, y, d\rangle = |N_x + 1 - x, y, \sigma_x(d)\rangle, \tag{3.13}$$
$$\mathcal{P}_y|x, y, d\rangle = |x, N_y + 1 - y, \sigma_y(d)\rangle, \tag{3.14}$$

where σ_x exchanges only R, L, and σ_y exchanges only U, D. Let

$$\mathcal{R} = \mathcal{P}_x\mathcal{P}_y, \qquad \mathcal{R}|x, y, d\rangle = |N_x + 1 - x, N_y + 1 - y, \bar{d}\rangle. \tag{3.15}$$

The rectangle and the moving shift obey

$$\boxed{[S, \mathcal{P}_x] = [S, \mathcal{P}_y] = [S, \mathcal{R}] = 0.} \tag{3.16}$$

These are real commuting involutions. For a nonsquare rectangle their geometric group is $D_2 = \{I, \mathcal{P}_x, \mathcal{P}_y, \mathcal{R}\}$, of order four. This is the geometric group of the rectangle, not the entire operator commutant of S. Repeated cycles and degeneracies allow additional shift-only conserved operators. The common-commutant formulation in Appendix B covers that broader question. The simultaneous parity projectors are

$$\Pi_{p_x p_y} = \frac{1}{4}(I + p_x\mathcal{P}_x)(I + p_y\mathcal{P}_y), \qquad p_x, p_y \in \{+1, -1\}. \tag{3.17}$$

The commutators hold at the wall as well as in the bulk. A reflected existing link remains existing, with its direction reflected. A reflected blocked link stays blocked, and reflection and opposite-direction reversal are compatible. Thus both orders of action agree on every basis state. Position reflection alone or direction permutation alone generally would not have this property.

Their ranks can be large. A one-dimensional group representation means that each symmetry acts by a number ± 1; it does *not* mean the entire symmetry sector has dimension one.

Since $\mathcal{R}$ has no fixed position–direction basis states, $\operatorname{Tr}\mathcal{R} = 0$. Its two projectors $\Pi_\pm = (I \pm \mathcal{R})/2$ have equal rank $2N$. For a square, diagonal reflection is available as well and the full geometric group is D_4. One must then check which of those operations also survive the coin.

The same geometry provides reversing operators:

$$P \in \{\mathcal{J}, \mathcal{P}_x\mathcal{J}, \mathcal{P}_y\mathcal{J}, \mathcal{R}\mathcal{J}\} \quad \Longrightarrow \quad PSP = S^{-1}. \tag{3.18}$$

The last operator reflects both position coordinates and leaves the direction label unchanged. These operators will enter the full-walk time-reversal test in Chapter 6.

3.5 Homogeneous bulk motion does not restore translation symmetry

The coin will be homogeneous, but the finite rectangle has walls. Translating a state by one site does not generally preserve the allowed set and its edge rule. The finite walk therefore has no automatic periodic translation symmetry or globally conserved plane-wave momentum.

Away from the walls, a plane wave can still be used to describe local bulk evolution. Its matrix is

$$U(k_x, k_y) = \Lambda(k_x, k_y)C, \qquad \Lambda = \operatorname{diag}(e^{-ik_x}, e^{ik_x}, e^{-ik_y}, e^{ik_y}). \tag{3.19}$$

The wall equations must then be imposed separately. The existence of this bulk matrix is not a finite-rectangle separability theorem.

Chapter 4

Coherent channels, envelopes, and the 16-cell map of a coin

4.1 The question of this chapter

The shift on the rectangle, with no coin yet, supports standing waves: complete sets of states that the shift keeps inside small, explicitly known subspaces (constructed in Sections 4.5 and 4.6 below). Grover's exact solution (Section 6.6) works because Grover's coin also keeps those subspaces. This chapter turns that observation into a test that can be applied to *any* coin, *before* the full walk is diagonalized:

Question

Given only the 4×4 coin matrix C, does the full walk $U = S(I_{\text{pos}} \otimes C)$ keep the standing waves of the rectangle, and if not, how much of each wave does it lose per step?

The answer requires three ingredients, developed in order:

1. a basis of the four internal states adapted to the walls, the *coherent channels* (Section 4.2);
2. the notion of the *envelope* of a state, and the rule that assigns an envelope to each coherent channel (Sections 4.3–4.6);
3. the observation that a coin can change the internal state but never the envelope. As a result, everything is decided by the sixteen entries of C written in the coherent basis (Sections 4.8–4.10).

Figure 4.1 summarizes the three steps. Its panels (a), (b) and (c) correspond to Sections 4.2, 4.4 and 4.10.

4.2 Two bases for the four internal states; the coherent channels

So far the internal states have been written in the *physical basis* $\{|R\rangle, |L\rangle, |U\rangle, |D\rangle\}$: each vector is a definite direction of motion. This basis is natural for the shift in the bulk, where each direction simply moves. At a wall it is not natural. Consider one row and only the horizontal directions:

- at the right wall the blocked move sends $|N_x, y, R\rangle$ to $|N_x, y, L\rangle$;
- at the left wall it sends $|1, y, L\rangle$ to $|1, y, R\rangle$.

On the pair space $\text{span}\{|R\rangle, |L\rangle\}$ the wall therefore acts as the exchange $R \leftrightarrow L$, which is the

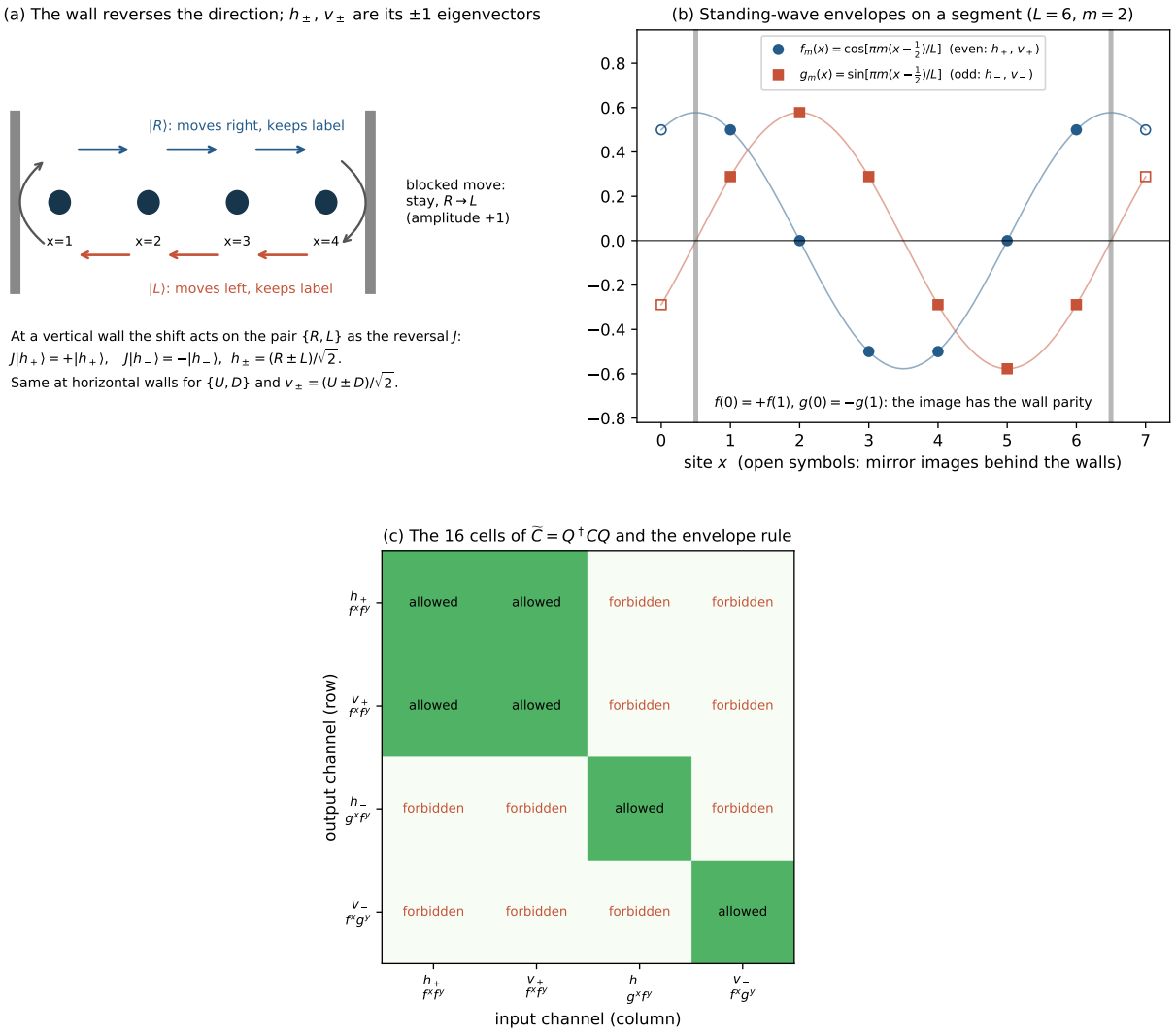


Figure 4.1: (a) A blocked move keeps the walker on its site and reverses its direction with amplitude +1. On the pair $\{R, L\}$ this is the exchange J; its eigenvectors are the coherent channels $h_\pm = (R \pm L)/\sqrt{2}$, with eigenvalues ±1, and likewise $v_\pm$ for $\{U, D\}$ at the horizontal walls. (b) The envelope of a channel along its own axis, on a segment of six sites ($m = 2$). A channel with wall eigenvalue +1 has a cosine envelope f_m, whose mirror value behind the wall (open symbol) has the same sign. A channel with eigenvalue −1 has a sine envelope g_m, whose mirror value has the opposite sign. (c) The coin matrix written in the coherent basis (h_+, v_+, h_-, v_-): rows are output channels, columns are input channels, and each label lists the envelope that the channel carries in a standing wave. Only the six cells joining channels with the same envelope are allowed.

horizontal block of the reversal J of Chapter 2. Neither $|R\rangle$ nor $|L\rangle$ is returned as itself; each is turned into the other.

The combinations that the wall returns as themselves, up to a sign, are the eigenvectors of the exchange. We define the four **coherent channels**

$$\boxed{|h_\pm\rangle = \frac{|R\rangle \pm |L\rangle}{\sqrt{2}}, \qquad |v_\pm\rangle = \frac{|U\rangle \pm |D\rangle}{\sqrt{2}}, \qquad J|h_\pm\rangle = \pm|h_\pm\rangle, \quad J|v_\pm\rangle = \pm|v_\pm\rangle.} \tag{4.1}$$

They form a second orthonormal basis of the same four-dimensional space $\mathbb{C}^4$, the *coherent basis*. "Coherent" refers to the fact that each channel is a superposition with fixed relative sign of two opposite directions, rather than a single direction. The letter h marks the horizontal pair and v the vertical pair. The sign $\pm$ is the channel's **wall parity**: the factor it acquires when it is reversed at a wall.

Two properties of the moving shift will be used below.

- A horizontal channel moves only along x, so it is blocked only by the two vertical walls $x = 1$ and $x = N_x$. Its wall parity therefore constrains its spatial profile along x, not along y. Likewise $v_\pm$ feel only the horizontal walls and are constrained only along y.
- The shift never mixes the horizontal pair with the vertical pair, and the walls never mix a $+$ channel with a $-$ channel. Any such mixing in the full walk must come from the coin.

4.3 The envelope of a state

A general walker state assigns an amplitude $\psi_d(\mathbf{r})$ to every site and internal state. Many important states have a simpler, *product* form: one fixed internal vector $|\chi\rangle$ multiplied by one spatial function $F(\mathbf{r})$,

$$|\Psi\rangle = \sum_{\mathbf{r}\in\Omega} F(\mathbf{r})|\mathbf{r}\rangle \otimes |\chi\rangle \equiv |F\rangle \otimes |\chi\rangle. \tag{4.2}$$

We call the spatial function F the **envelope** of the state and $|\chi\rangle$ its internal state. For example, $|f^x_m f^y_n\rangle \otimes |h_+\rangle$ has envelope $F(x, y) = f_m(x)f_n(y)$ and internal state h_+: at every site the walker is in the same superposition $(R + L)/\sqrt{2}$, with the amplitude modulated in space by F. An envelope is defined only for product states. A general state is a sum of such terms, each with its own envelope and channel.

The point of this definition is that the coin and the shift treat the two factors differently:

- the coin $I_{\text{pos}} \otimes C$ changes the internal state and leaves the envelope untouched: $|F\rangle \otimes |\chi\rangle \mapsto |F\rangle \otimes C|\chi\rangle$;
- the shift moves amplitude between sites, so it changes the envelope. At a wall it also multiplies the channel by its wall parity.

4.4 Panel (b): the wall parity fixes the envelope of each channel

Which envelopes are compatible with the walls? Consider a horizontal channel of parity ± 1 on a row of $L = N_x$ sites. The stay-and-reverse rule says that amplitude arriving at the wall comes back from the same site, multiplied by the parity. This is exactly what one obtains by continuing the envelope to a mirror site behind each wall, at $x = 0$ and $x = L + 1$, with

- the *same sign* for parity $+1$: $F(0) = F(1)$ and $F(L + 1) = F(L)$;
- the *opposite sign* for parity -1: $F(0) = -F(1)$ and $F(L + 1) = -F(L)$.

The mirror plane lies half a site beyond the last allowed site. The standing waves that satisfy these conditions are the half-site-shifted cosines f_m (parity $+1$) and sines g_m (parity -1) defined precisely in Eq. (4.3) below and drawn in Fig. 4.1(b). The same holds for a vertical channel along y.

Each channel's parity fixes the envelope factor along its own axis. The factor along the other axis is fixed by the requirement, explained in Section 4.6, that the four channels form one invariant subspace of the shift. The result, used throughout the notes, is:

Coherent channel	Moves along	Wall parity	Envelope in the mode (m, n)
h_+	x	$+1$	$f_m(x)\, f_n(y)$
v_+	y	$+1$	$f_m(x)\, f_n(y)$
h_-	x	-1	$g_m(x)\, f_n(y)$
v_-	y	-1	$f_m(x)\, g_n(y)$

Two channels share an envelope only in the case of h_+ and v_+. h_- and v_- have the same parity, but their sines run along different axes, so their envelopes differ.

4.5 The standing waves of the reflecting shift

Recall the coherent channels of Eq. (4.1). On a segment of L sites define normalized profiles

$$f_m^{(L)}(j) = \sqrt{\frac{2-\delta_{m0}}{L}} \cos\left[\frac{\pi m}{L}(j-\tfrac{1}{2})\right], \qquad m = 0, \ldots, L-1, \tag{4.3}$$

$$g_m^{(L)}(j) = \sqrt{\frac{2-\delta_{mL}}{L}} \sin\left[\frac{\pi m}{L}(j-\tfrac{1}{2})\right], \qquad m = 1, \ldots, L. \tag{4.4}$$

Each family separately is an orthonormal basis on the L sites. The half-site offset makes its even or odd continuation compatible with the stay-and-reverse boundary rule.

For $1 \le m < N_x$, with $k_m = \pi m/N_x$, the horizontal shift satisfies the verified identities

$$S_x|f_m^x; h_+\rangle = \cos k_m |f_m^x; h_+\rangle + \sin k_m |g_m^x; h_-\rangle, \tag{4.5}$$

$$S_x|g_m^x; h_-\rangle = -\sin k_m |f_m^x; h_+\rangle + \cos k_m |g_m^x; h_-\rangle. \tag{4.6}$$

Here $f_m^x = f_m^{(N_x)}$. Trigonometric addition formulas verify the interior relations, and the even/odd continuations verify both endpoints. There are also endpoint modes

$$S_x|f_0^x; h_+\rangle = |f_0^x; h_+\rangle, \qquad S_x|g_{N_x}^x; h_-\rangle = -|g_{N_x}^x; h_-\rangle. \tag{4.7}$$

The vertical formulas replace x, h, N_x, m by y, v, N_y, n.

Equation (4.6) says that the shift can mix a cosine component and a sine component while preserving their common mode index. Conservation of a mode label is not the absence of evolution.

4.6 The complete mode sectors and their projectors

For $0 \le m < N_x$, $0 \le n < N_y$, group the following orthonormal states:

$$|a_{mn}\rangle = |f^x_m\rangle \otimes |f^y_n\rangle \otimes |h_+\rangle, \tag{4.8}$$
$$|b_{mn}\rangle = |f^x_m\rangle \otimes |f^y_n\rangle \otimes |v_+\rangle, \tag{4.9}$$
$$|c_{mn}\rangle = |g^x_m\rangle \otimes |f^y_n\rangle \otimes |h_-\rangle, \qquad m > 0, \tag{4.10}$$
$$|d_{mn}\rangle = |f^x_m\rangle \otimes |g^y_n\rangle \otimes |v_-\rangle, \qquad n > 0. \tag{4.11}$$

Their span is $\mathcal{K}_{mn}$, with dimension $r_{mn} = 2 + \mathbf{1}_{m>0} + \mathbf{1}_{n>0}$. Zero-index sine terms are omitted. The associated projector is explicitly

$$\boxed{P_{mn} = |a_{mn}\rangle\langle a_{mn}| + |b_{mn}\rangle\langle b_{mn}| + \mathbf{1}_{m>0}|c_{mn}\rangle\langle c_{mn}| + \mathbf{1}_{n>0}|d_{mn}\rangle\langle d_{mn}|.} \tag{4.12}$$

Terms whose vectors are absent are simply left out of this sum.

To complete the basis, include the terminal singletons

$$\mathcal{E}^H_n = \mathrm{span}\{|g^x_{N_x} f^y_n; h_-\rangle\}, \qquad 0 \le n < N_y, \tag{4.13}$$
$$\mathcal{E}^V_m = \mathrm{span}\{|f^x_m g^y_{N_y}; v_-\rangle\}, \qquad 0 \le m < N_x. \tag{4.14}$$

Their projectors are the outer products of the displayed normalized vectors. All these spaces are mutually orthogonal and

$$\sum_{m,n} r_{mn} + N_x + N_y = 4N_xN_y. \tag{4.15}$$

They form a complete decomposition, denoted $\mathscr{P}_{\mathrm{SW}}$. The shift preserves every sector:

$$[S, P_a] = 0 \qquad (P_a \in \mathscr{P}_{\mathrm{SW}}). \tag{4.16}$$

The index a collectively denotes (m,n), (H,n), or (V,m). Every P_a acts on position *and* coin and has matrix size $D \times D$. It is not a local 4×4 projector.

Each sector is the smallest shift-invariant space containing the two symmetric states a_{mn}, b_{mn}: the shift rotates a_{mn} into c_{mn} along x and b_{mn} into d_{mn} along y (Eq. (4.6)), never across. This is why the four channels of one sector carry the four envelopes of Section 4.4. These modes are constructed before selecting C and without diagonalizing U. Whether a coin preserves them is a subsequent question. The row/column decomposition and this standing-wave decomposition are different useful organizations of the same shift. They need not survive the same coin.

4.7 The simplest example: the identity coin

Before any nontrivial coin is introduced, it helps to see the cycles and the standing waves at work in the one case where nothing else happens. Take $C = I_4$. Then

$$U = S(I_{\mathrm{pos}} \otimes I_4) = S : \tag{4.17}$$

the walk *is* the shift. This is the precise meaning of "a walk without a coin" in this model. The four internal states are still present, because the shift needs the direction label to know where to move and what to reverse at a wall. What is absent is any mixing of the directions. A walk with no internal space at all would be a different model. A global phase, $C = e^{i\phi} I_4$, is equally trivial: it only rotates every eigenphase by ϕ.

Cycles. Since nothing mixes the directions, a walker that starts in $|x, y, R\rangle$ moves right along its row, is reversed at the wall, returns as L, is reversed again at the other wall, and so on

(Fig. 4.2(a)). Every row is one closed cycle of $2N_x$ basis states and every column one of $2N_y$. A state never leaves its cycle, so the dynamics is a set of independent one-dimensional rotations. Fourier transformation along each cycle gives the complete spectrum:

$$\operatorname{spec} U = \left\{e^{i\pi k/N_x}\right\}_{k=0}^{2N_x-1} \text{ each } N_y \text{ times} \cup \left\{e^{i\pi k/N_y}\right\}_{k=0}^{2N_y-1} \text{ each } N_x \text{ times}, \qquad U^{2\,\mathrm{lcm}(N_x,N_y)} = I. \tag{4.18}$$

Exact result. These are the shift cycles of Chapter 3. **Computed for this revision.** Direct diagonalization agrees with Eq. (4.18) to 2×10^{-15} on 7×5 and 8×6. The smallest periods found, 70 and 48, equal $2\,\mathrm{lcm}(N_x,N_y)$. The identical copies of every row and every column make this spectrum massively degenerate, so level statistics are meaningless here.

Standing waves inside a cycle. The same dynamics can be read in the standing-wave basis. On one row, the shift turns the cosine-envelope state $f_m h_+$ into the sine-envelope state $g_m h_-$ and back, by the angle $\pi m/N_x$ per step (Eq. (4.6)). The weight never leaves this pair, as Fig. 4.2(c) shows. The eigenvalues $e^{\pm i\pi m/N_x}$ of this two-dimensional rotation, together with $+1$ for $f_0 h_+$ and -1 for $g_{N_x} h_-$, are exactly the $2N_x$ roots of unity of one row cycle. The standing waves are therefore not something added to the shift: they are another basis for the same cycles, one in which the wall rule is built into the envelopes.

Running versus standing waves. With $C = I$ the sector $\mathcal{K}_{mn}$ splits further, into an x-pair (a_{mn}, c_{mn}) and a y-pair (b_{mn}, d_{mn}) that never talk to each other. The eigenvector $(a_{mn} - ic_{mn})/\sqrt{2}$ of the x-pair is a wave running around the row cycle. Its spatial density is uniform along x, because $f_m^2 + g_m^2$ is constant for a nonterminal index; only the y standing wave survives (Fig. 4.3, bottom left). A coin that couples h_+ with v_+, such as Grover, joins the two pairs into one four-dimensional block. Its eigenstates then carry a genuinely two-dimensional standing-wave pattern (Fig. 4.3, bottom middle). Both displayed states were checked to be exact eigenvectors of the full walk, with residual below 10^{-15}.

In the language of the rest of this chapter, the identity coin is the member $C_+ = I_2$, $\alpha = \beta = 0$ of the exactly solvable family of Section 4.10. Its 16-cell square has weight only on the diagonal, in allowed cells. It also belongs to the other three solvable families of Section 4.11: it is monomial, it is block diagonal by axis, and it is trivially its own gauge image. It is the most degenerate integrable walk of the rectangle.

4.8 Why a 4×4 square decides compatibility

We now combine the two previous steps. Take one standing-wave state of a sector $\mathcal{K}_{mn}$: a product state with coherent channel j and the envelope E_j that the table of Section 4.4 assigns to channel j. Apply the coin. By Section 4.3, the envelope is untouched and only the internal state changes:

$$(I_{\mathrm{pos}} \otimes C)\,|E_j\rangle \otimes |j\rangle = \sum_i \widetilde{C}_{ij}\; |E_j\rangle \otimes |i\rangle, \qquad \widetilde{C}_{ij} = \langle i|C|j\rangle, \tag{4.19}$$

where i, j run over the four coherent channels. Every output term keeps the *input's* envelope E_j but now sits in channel i. Whether the result is still inside $\mathcal{K}_{mn}$ therefore depends on only two things:

- the 4×4 numbers $\widetilde{C}_{ij}$, which say how much amplitude goes from channel j to channel i;
- whether the envelope E_j is the one that $\mathcal{K}_{mn}$ requires for channel i, that is whether $E_i = E_j$.

The shift needs no further analysis, because it already keeps every $\mathcal{K}_{mn}$ (Section 4.6).

So the whole question reduces to a table with four inputs and four outputs: the coin matrix itself, written as a square of $4\times4 = 16$ cells. This is the **16-cell square** used for every coin in these

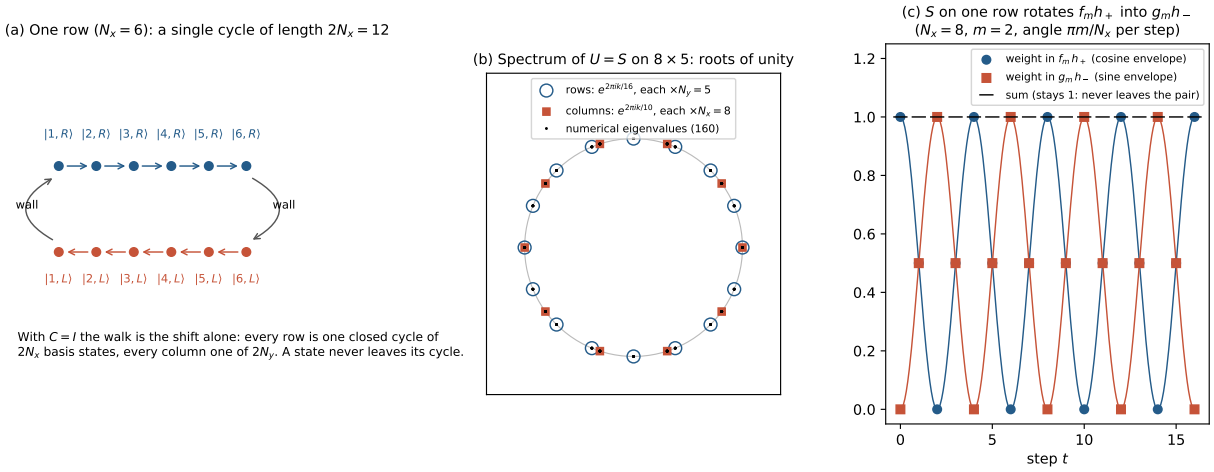


Figure 4.2: The identity coin, $U = S$. (a) One row of six sites forms a single cycle of twelve basis states: R states move right, the wall turns R into L, L states move left, and the other wall closes the cycle. (b) Spectrum on 8×5: the 16th roots of unity from the rows, each repeated $N_y = 5$ times, and the 10th roots from the columns, each repeated $N_x = 8$ times. Black dots are the numerically computed eigenvalues. (c) Starting from the cosine-envelope standing wave $f_2 h_+$ on a row of eight sites, the shift alone rotates the weight into the sine-envelope state $g_2 h_-$ and back, by $\pi/4$ per step. The sum of the two weights stays equal to one.

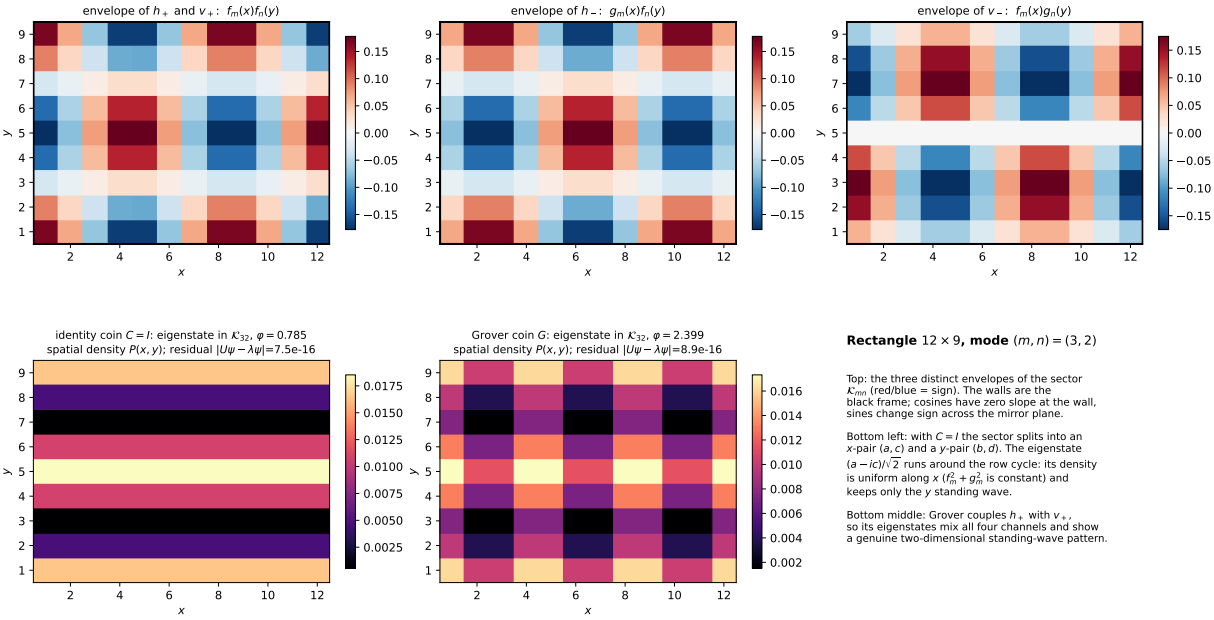


Figure 4.3: Standing waves on a 12×9 rectangle, mode $(m, n) = (3, 2)$. Top: the three distinct envelopes of the sector $\mathcal{K}_{32}$ (colour = value and sign). h_+ and v_+ share $f_m(x)f_n(y)$; h_- carries $g_m(x)f_n(y)$ and v_- carries $f_m(x)g_n(y)$. The black frame marks the walls: a cosine factor is flat across the mirror plane, and a sine factor changes sign across it. Bottom: spatial density $P(x, y) = \sum_d |\psi_d(x, y)|^2$ of one exact eigenstate in this sector. Left, identity coin: a wave running around the row cycle, uniform along x. Middle, Grover: the four channels are coupled and the density is a two-dimensional standing-wave pattern.

notes. It is not an extra construction or an auxiliary 4×4 object: it is the coin matrix, displayed in a basis chosen to make the envelope question readable.

4.9 Which basis may be used for the square?

Any orthonormal basis gives a valid square. A 4×4 unitary can be written in any orthonormal basis $\{|e_1\rangle, \ldots, |e_4\rangle\}$ of $\mathbb{C}^4$. If the basis vectors are the columns of a unitary matrix Q, the entries are

$$\widetilde{C} = Q^\dagger C Q, \qquad \widetilde{C}_{ij} = \langle e_i | C | e_j \rangle.$$

All such squares describe the same operator, with the same eigenvalues and the same walk. In every basis the cell (i, j) is the amplitude from input e_j to output e_i. Rows are outputs and columns are inputs. Because C is unitary every column is a normalized vector, so for a fixed input the numbers $|\widetilde{C}_{ij}|^2$ are the probabilities of the four outputs.

Different bases answer different questions. Two bases are used in these notes, and Fig. 4.4 shows the Grover coin in both:

Basis	Columns of Q	Question answered cell by cell
Physical (R, L, U, D)	$Q = I_4$, so the square is C itself	Does a definite direction of motion split into several directions? (splitting measures of Chapter 5; routing)
Coherent (h_+, v_+, h_-, v_-)	Eq. (4.20)	Does the coin keep the envelopes of the standing waves?

Only the coherent basis carries the envelope rule. The allowed/forbidden reading of Section 4.10 needs every basis vector to have *one definite envelope* in the standing-wave sectors. The coherent channels have this property (table of Section 4.4). The physical directions do not: in a standing wave the amplitude on R is a sum of an h_+ part with a cosine envelope and an h_- part with a sine envelope. A physical-basis cell therefore mixes allowed and forbidden contributions and cannot be labeled either way. The mask of Fig. 4.1(c) applies to the coherent square only.

What may still be chosen freely.

- *The order of the four channels.* We use (h_+, v_+, h_-, v_-) because it makes the allowed cells a single 2×2 block followed by two diagonal cells. Another order permutes rows and columns together and moves the allowed cells, but does not change which pairs are allowed.
- *Phases of the basis vectors.* These also leave the envelope rule unchanged.

Changing the coin itself by phases, $C \to DCD$, is a different operation, a *gauge transformation* of the walk. It is discussed in Section 4.11.

With the coherent channels as columns, in the physical order (R, L, U, D),

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \qquad \widetilde{C} = Q^\dagger C Q. \tag{4.20}$$

For Grover, Fig. 4.4 shows that the two squares look completely different. Every physical cell has modulus $1/2$: each direction splits into all four. The coherent square is a signed permutation: $h_+ \leftrightarrow v_+$, $h_- \to -h_-$, $v_- \to -v_-$. Equal probabilities in the physical square can hide a very sparse coherent square, because the relative phases matter.

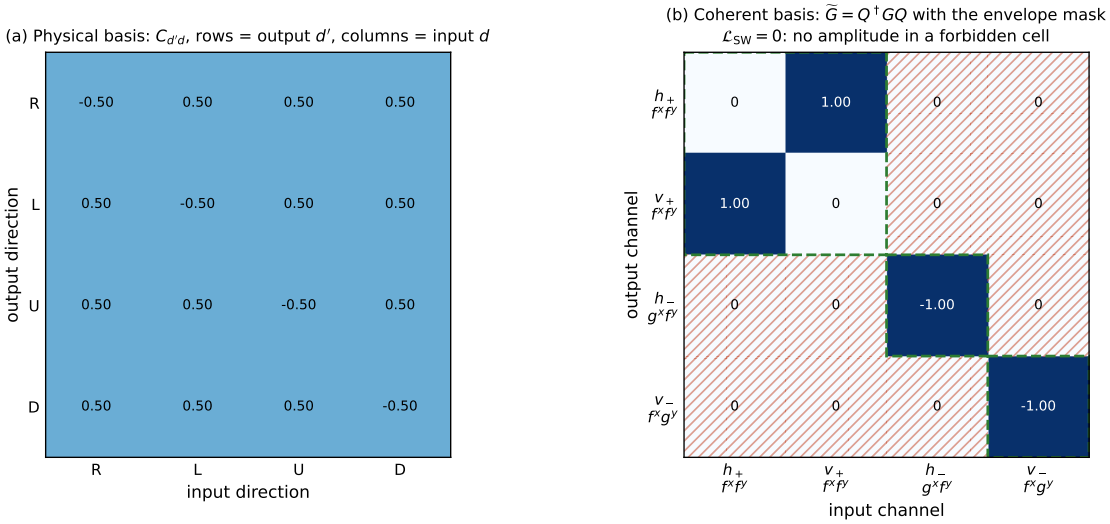


Figure 4.4: The Grover coin as a 16-cell square in two bases. (a) Physical basis (R, L, U, D): the cell in row d' and column d is $C_{d'd}$, the amplitude from input direction d to output direction d'; colour is its modulus. No envelope mask applies in this basis. (b) Coherent basis (h_+, v_+, h_-, v_-): the cell in row i and column j is $\widetilde{G}_{ij} = \langle i|G|j\rangle$. The envelope mask is superimposed: hatched cells are forbidden, and the dashed green boxes are the allowed $2+1+1$ blocks. Grover puts no amplitude in a forbidden cell.

4.10 Panel (c): allowed and forbidden cells

The rule. By Eq. (4.19), the amplitude $\widetilde{C}_{ij}$ keeps the input's envelope E_j while moving it to channel i. It stays inside the sector $\mathcal{K}_{mn}$ exactly when $E_i = E_j$. Otherwise it is orthogonal to the channel-i state of the sector, because $\langle f_m, g_m\rangle = 0$ for every nonterminal index, and it leaves $\mathcal{K}_{mn}$.

The sixteen cells. Reading the envelopes from the table of Section 4.4:

- *allowed* (6 cells):
  - the 2×2 block $\{h_+, v_+\} \to \{h_+, v_+\}$, since both channels carry $f_m f_n$;
  - the diagonal cells $h_- \to h_-$ and $v_- \to v_-$;
- *forbidden* (10 cells):
  - the eight cells that change the wall parity $(+ \leftrightarrow -)$;
  - the two cells $h_- \leftrightarrow v_-$, which keep the parity but exchange the envelope $g_m f_n$ for $f_m g_n$.

In the order (h_+, v_+, h_-, v_-) the allowed cells form the block-diagonal pattern $2{+}1{+}1$ of Fig. 4.1(c).

One number per square. The forbidden weight is

$$\boxed{\mathcal{L}_{\mathrm{SW}}(C) = \frac{1}{4} \sum_{(i,j)\ \text{forbidden}} |\widetilde{C}_{ij}|^2 = 1 - \frac{1}{4}\Big(\sum_{a,b=1}^{2} |\widetilde{C}_{ab}|^2 + |\widetilde{C}_{33}|^2 + |\widetilde{C}_{44}|^2 \Big).} \tag{4.21}$$

- For each input channel j, the sum of $|\widetilde{C}_{ij}|^2$ over its forbidden cells is the probability that the coin sends that channel out of its standing-wave sector.
- The factor $1/4$ averages over the four input channels.
- Since the shift keeps every sector, $\mathcal{L}_{\mathrm{SW}}$ equals the mean probability per step that the full walk leaves the standing-wave sectors. This is the exact full-space identity of Section 6.4,

Eq. (6.10).

The exactly solvable family. $\mathcal{L}_{\mathrm{SW}} = 0$ holds if and only if all forbidden cells vanish, that is, if the coherent square is block diagonal:

$$\widetilde{C} = C_+ \oplus e^{i\alpha} \oplus e^{i\beta}, \qquad C_+ \in U(2). \tag{4.22}$$

Then every $\mathcal{K}_{mn}$ is invariant under the full walk, which is solved by blocks of dimension at most four. Combining the shift rotation (4.6) with the coin, the restricted step in the order $(a_{mn}, b_{mn}, c_{mn}, d_{mn})$ is, for $m, n > 0$,

$$\boxed{U_{mn} = S_{mn}\left(C_+ \oplus e^{i\alpha} \oplus e^{i\beta}\right), \qquad S_{mn} = \begin{pmatrix} c_x & 0 & -s_x & 0 \\ 0 & c_y & 0 & -s_y \\ s_x & 0 & c_x & 0 \\ 0 & s_y & 0 & c_y \end{pmatrix},} \tag{4.23}$$

with $c_x = \cos(\pi m/N_x)$, $s_x = \sin(\pi m/N_x)$, and similarly c_y, s_y. Rows and columns of absent vectors are deleted when $m = 0$ or $n = 0$, and at $(0, 0)$ the block is C_+ itself. The N_y horizontal terminal singletons have eigenvalue $-e^{i\alpha}$, and the N_x vertical ones have eigenvalue $-e^{i\beta}$.

For $C_+ = \sigma_x$ and $\alpha = \beta = \pi$ this reproduces the Grover blocks of Section 6.6. In a numerical check for this revision, the union of these block spectra matched direct diagonalization of the full walk to 4.6×10^{-15} on 7×5 and 7.8×10^{-15} on 8×6. After a gauge transformation, the same formula also gives the compass Fourier walk (Section 6.7).

4.11 What the square can and cannot decide

The mask tests one family of geometric modes. A zero forbidden weight is *sufficient* for an exact block solution. A nonzero weight is not a proof of non-integrability, for three demonstrated reasons, each with its own fingerprint in the squares.

- **Gauge.** The walks built from C and from DCD, $D = \mathrm{diag}(e^{ia}, e^{-ia}, e^{ib}, e^{-ib})$, are unitarily equivalent (Appendix A). The equivalence maps the standing waves onto phase-modulated versions of themselves, so a coin may fill forbidden cells while its gauge image does not. The gauge-adjusted test is $\mathcal{L}^*_{\mathrm{SW}} = \min_D \mathcal{L}_{\mathrm{SW}}(DCD)$.
- **Routing.** A monomial coin, with one nonzero cell per column of the *physical* square, is solved by deterministic cycles whatever its coherent square looks like.
- **Independent axes.** A coin that never mixes $\{R, L\}$ with $\{U, D\}$ is solved row by row and column by column.

Reduction family	Fingerprint	Example
Fixed standing waves	Coherent square: no weight in forbidden cells	Grover G
Gauge images of standing waves	Coherent square of DCD: no weight in forbidden cells	Compass Fourier F_c: $\mathcal{L}_{\mathrm{SW}} = \frac{1}{2}$, $\mathcal{L}^*_{\mathrm{SW}} = 0$
Deterministic routing	Physical square: one nonzero cell per column	Permutation P: $\mathcal{L}_{\mathrm{SW}} = \mathcal{L}^*_{\mathrm{SW}} = \frac{1}{2}$
Independent rows and columns	Physical square block diagonal in $\{R, L\} \mid \{U, D\}$; coherent square block diagonal in $\{h_+, h_-\} \mid \{v_+, v_-\}$	$C_H \oplus C_V$, e.g. Hadamard per axis

The table also shows why both bases are needed. Two of the fingerprints are read in the coherent basis and two in the physical basis. These four families have not been proved exhaustive, so a coin outside all of them still has an open status.

4.12 What can already be said about coin compatibility?

If $\mathcal{G}$ is one of the geometric operations commuting with S, then

$$[U, \mathcal{G}] = S[\mathcal{C}, \mathcal{G}]. \tag{4.24}$$

For a homogeneous coin, the only remaining test is its commutator with the direction permutation in $\mathcal{G}$. In particular,

Condition on the coin	Surviving full-walk geometric symmetry
$[C, \sigma_x] = 0$	$\mathcal{P}_x$
$[C, \sigma_y] = 0$	$\mathcal{P}_y$
$[C, J] = 0$	$\mathcal{R}$
$[C, \tau] = 0$, on a square	Diagonal reflection

Likewise, because each mode projector commutes with S,

$$[U, P_a] = 0 \quad \Longleftrightarrow \quad [\mathcal{C}, P_a] = 0. \tag{4.25}$$

This is the bridge to the coin: the geometry supplies candidate structures, and the coin determines which remain conserved.

The rectangle supplies several levels of structure

The shift has individual row/column cycles, geometric reflection parities, and complete standing-wave sectors. These are different labels. A coin may destroy one decomposition while preserving another. The rectangle alone does not determine the symmetries or solvability of the full Floquet operator.

4.13 Inventory before the coin is introduced

Structure	Established for the shift	Question left for the coin
Row and column projectors	Independent cycles of lengths $2N_x, 2N_y$	Does it remain within each axis?
Geometric operations	$\mathcal{P}_x, \mathcal{P}_y, \mathcal{R}$ commute with S	Does it commute with their direction permutations?
Standing-wave projectors	Each explicit sector is invariant	Does it keep coin vectors and spatial envelopes compatible?
16-cell coherent square	Allowed cells fixed by wall parity and envelope	How much weight does the coin put in the ten forbidden cells?
Reversing operations	They turn S into S^{-1}	Does the transpose condition hold for full-walk time reversal?

These are distinct structures; the geometric group is not asserted to exhaust the shift's commutant. The existing advanced constructions, including uniform and checkerboard subspaces, are retained in Appendix B.

Chapter 5

The coins as local four-dimensional operators

5.1 How the coins are displayed

Every coin is given as an explicit matrix in the physical order (R, L, U, D) and then redrawn as its 16-cell coherent square $\widetilde{C} = Q^\dagger C Q$ with the envelope mask of Chapter 4. Figure 5.1 collects the unperturbed coins and Fig. 5.2 the two deformed families. The numbers in the panel titles are the fixed-mode forbidden weight $\mathcal{L}_{\mathrm{SW}}$ and, where computed, its gauge minimum $\mathcal{L}^*_{\mathrm{SW}}$. A block in the local matrix does not by itself preserve a full-space mode projector; the full walks are examined in Chapter 6.

5.2 Three questions that a "mixing measure" can ask

The physical direction basis is fixed. For a definite input direction d, the probability to obtain d' after the coin is

$$p(d'|d) = |C_{d'd}|^2. \tag{5.1}$$

These probabilities do not contain the relative phases of the amplitudes. Two distinct questions are whether an *individual direction splits* and whether a *pair label is redistributed*. A third question, coupling between spatial modes of the full walk, will be addressed in Chapter 6.

5.2.1 Individual-direction splitting

The normalized output entropy, averaged over the four definite input directions, is

$$\boxed{\mathcal{M}_{\mathrm{split}}(C) = -\frac{1}{4\ln 4}\sum_{d,d'} |C_{d'd}|^2 \ln |C_{d'd}|^2.} \tag{5.2}$$

Use $0 \ln 0 = 0$. It lies in $[0, 1]$. It is zero exactly when each column has a single nonzero entry. Unitarity then forces a permutation with phases, called a *monomial* coin. It equals one when every column has four probabilities $1/4$.

A second diagnostic, already used in the project, is

$$\mathcal{L}_{\mathrm{route}}(C) = 1 - \frac{1}{4}\max_{\pi \in S_4} \sum_{d} |C_{\pi(d),d}|^2. \tag{5.3}$$

It measures departure from the best deterministic routing. Its zero has the same monomial condition, although its nonzero values use a different scale from the entropy.

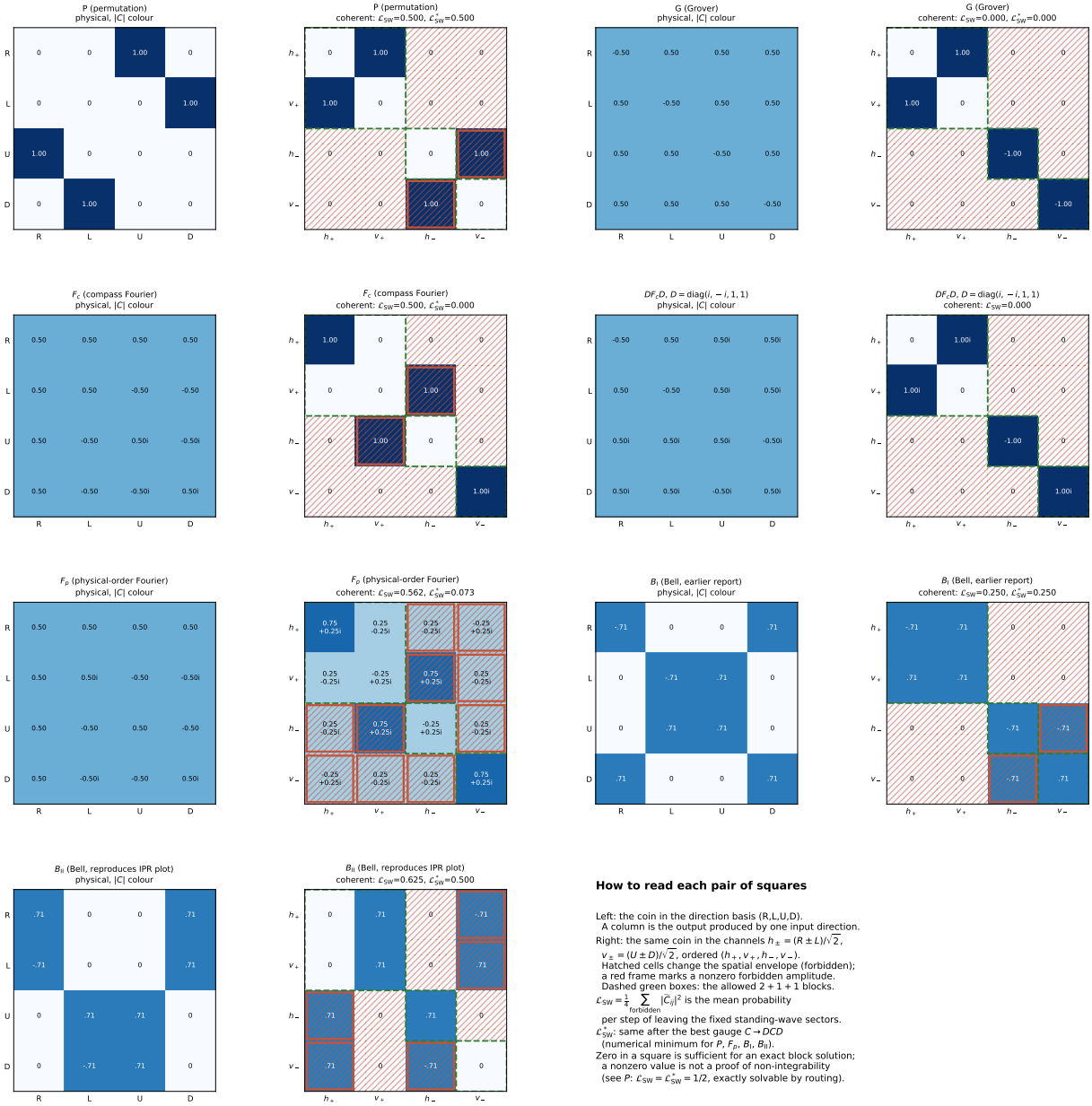


Figure 5.1: The unperturbed coins, each as a physical square (left of each pair) and as a coherent square with the envelope mask (right). Red frames mark nonzero amplitude in a forbidden cell. DF_cD is the gauge image of the compass Fourier coin with $D = \mathrm{diag}(i, -i, 1, 1)$: its forbidden cells are empty. $\mathcal{L}^*_{\mathrm{SW}}$ is exact for G and F_c (zero) and a numerical minimum (an upper bound) for P, F_p, B_{I} and B_{II}.

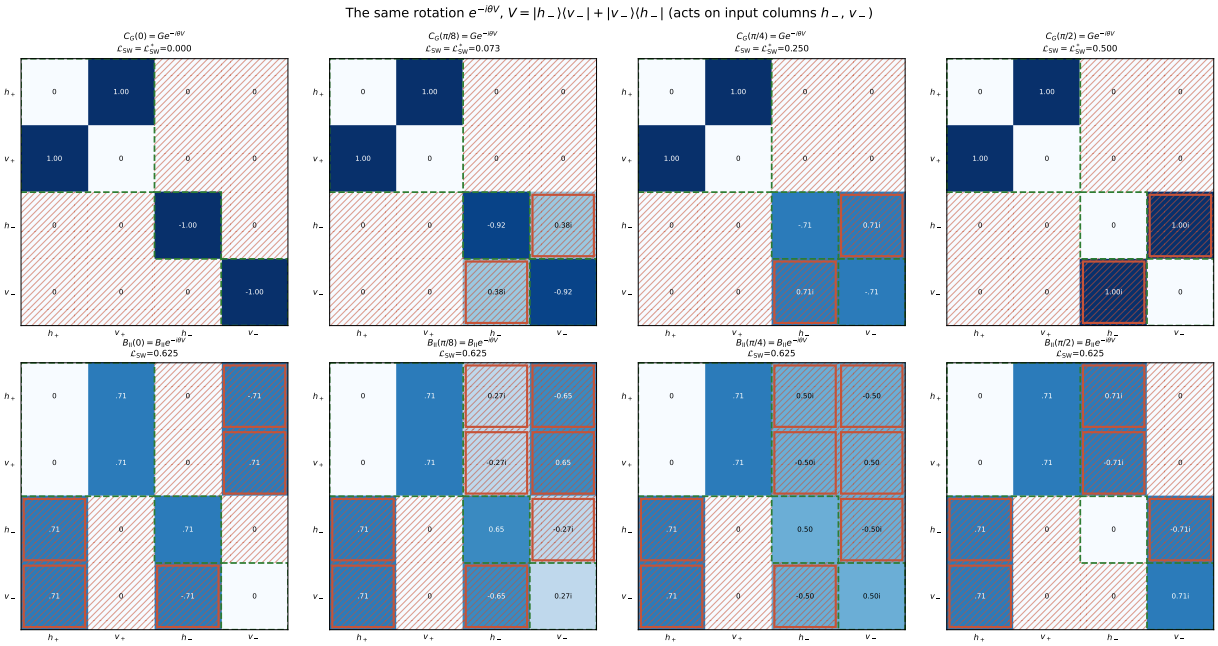


Figure 5.2: The same rotation $e^{-i\theta V}$ applied to Grover (top) and to B_{II} (bottom), at $\theta = 0, \pi/8, \pi/4, \pi/2$. Right multiplication acts on the input *columns* h_-, v_- only. For Grover it moves weight from the allowed diagonal cells $(h_-, h_-), (v_-, v_-)$ into the forbidden cells $(h_-, v_-), (v_-, h_-)$; for B_{II} it only redistributes weight that was already partly forbidden, and $\mathcal{L}_{\mathrm{SW}} = 5/8$ for every θ (Section 5.9).

5.2.2 Mixing a specified pair label

There are three unordered partitions of four directions into two pairs:

$$q_1 = \mathrm{diag}(1, 1, -1, -1), \qquad \{R, L\} \mid \{U, D\}, \tag{5.4}$$

$$q_2 = \mathrm{diag}(1, -1, 1, -1), \qquad \{R, U\} \mid \{L, D\}, \tag{5.5}$$

$$q_3 = \mathrm{diag}(1, -1, -1, 1), \qquad \{R, D\} \mid \{L, U\}. \tag{5.6}$$

For a pair A,

$$q_A = \sum_{d \in A} |d\rangle\langle d| - \sum_{d \notin A} |d\rangle\langle d|. \tag{5.7}$$

It assigns signs rather than exchanging directions: $q_A = q_A^\dagger$, $q_A^2 = I_4$, $\mathrm{Tr}\, q_A = 0$. Its two eigenspaces identify the groups, and reversing its sign gives the same unordered split. The original project definition is

$$\boxed{\mathcal{M}_q(C) = \frac{1}{8} \min\{\|[C, q]\|_F^2 , \|\{C, q\}\|_F^2\}.} \tag{5.8}$$

The Frobenius norm is $\|A\|_F^2 = \mathrm{Tr}(A^\dagger A)$, and $\{C, q\} = Cq + qC$. A zero means the coin preserves *or* swaps these two groups. It does not mean each individual direction remains unsplit.

For $P_\pm = (I \pm q)/2$, the transition probabilities for inputs $P_\pm/2$ form

$$T_q = \frac{1}{2} \begin{pmatrix} \|P_+ C P_+\|_F^2 & \|P_+ C P_-\|_F^2 \\ \|P_- C P_+\|_F^2 & \|P_- C P_-\|_F^2 \end{pmatrix} = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}. \tag{5.9}$$

The special symmetric form follows from unitarity and the equal ranks of the groups. Explicitly, $p = \frac{1}{2} \|P_+ C P_+\|_F^2$. Equivalently,

$$\|[C, q]\|_F^2 = 16(1-p), \qquad \|\{C, q\}\|_F^2 = 16p \tag{5.10}$$

These norm formulas make the normalization explicit: when both outputs have probability 1/2, the two squared norms are eight and the score is one. Exact preservation has $p = 1$; exact swapping has $p = 0$; both give zero, so $\mathcal{M}_q = 2\min(p, 1-p)$. The input here is an equal incoherent mixture inside one pair, not every individual pure direction state. This is why a deterministic permutation can have a nonzero pair score.

The two summaries

$$\mathcal{M}_{\text{dir}} = \min_{\alpha=1,2,3} \mathcal{M}_{q_\alpha}, \qquad \overline{\mathcal{M}} = \frac{1}{3}\sum_{\alpha=1}^{3} \mathcal{M}_{q_\alpha} \tag{5.11}$$

answer different questions. A zero of the minimum asserts that at least one of the three pair partitions is preserved or swapped.

5.2.3 The 24 permutations versus the three pair labels

A permutation satisfies $P_\pi|d\rangle = |\pi(d)\rangle$. There are $4! = 24$ relabelings but only $\binom{4}{2}/2 = 3$ unordered splits into two pairs:

$$P_\pi q_A P_\pi^\dagger = q_{\pi(A)}.$$

The permutations rearrange the partitions. The q_α are observables used to test pair preservation or swapping, not a list of asserted coin symmetries. Grover's 24 equations $P_\pi G P_\pi^\dagger = G$ express invariance under relabeling; they do not imply zero $\mathcal{M}_q$. A permutation matrix can also be chosen as the coin itself, a different role from testing the symmetry of another coin.

5.2.4 The proposed horizontal/vertical entropy

Write $C = \left(\begin{smallmatrix} C_{HH} & C_{HV} \\ C_{VH} & C_{VV} \end{smallmatrix}\right)$, grouping R, L and U, D. With the standard norm $\|C\|_F^2 = 4$, define

$$p = \frac{\|C_{HH}\|_F^2 + \|C_{VV}\|_F^2}{4}, \qquad \mathcal{M}_{\text{axis}} = h_2(p) = -\frac{p\ln p + (1-p)\ln(1-p)}{\ln 2}. \tag{5.12}$$

This is a valid entropy for the axis output conditioned on the coarse axis input, with an incoherent uniform state inside each group. It is an entropy version of the original q_1 test:

$$\boxed{\mathcal{M}_{\text{axis}} = h_2(\mathcal{M}_{q_1}/2).} \tag{5.13}$$

It introduces no new zero condition. Always retaining the axis and always swapping it both give zero.

For example, the permutation exchanging L, U and fixing R, D has $p = 1/2$ and maximal axis entropy, although no input direction splits. This does not invalidate the entropy; it identifies the coarse question it answers.

5.3 The identity coin

The trivial coin $C = I_4$ is the reference point for all others. In every basis its square is the identity, so no direction splits and no channel changes:

$$\mathcal{M}_{\text{split}} = 0, \qquad (\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3}) = (0,0,0), \qquad \mathcal{L}_{\text{SW}} = \mathcal{L}_{\text{SW}}^* = 0, \qquad [I, J] = 0. \tag{5.14}$$

Its full walk is the shift itself, whose cycles, spectrum and standing waves were made explicit in Section 4.7. Every other coin in this chapter can be read as a specific way of mixing the otherwise independent row and column cycles.

5.4 One explicit permutation coin

Take

$$\boxed{P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad R \leftrightarrow U, \quad L \leftrightarrow D.} \tag{5.15}$$

It is a real symmetric unitary with $P^2 = I$ and $[P, J] = 0$. Every input direction has a unique output, so

$$\mathcal{M}_{\text{split}}(P) = \mathcal{L}_{\text{route}}(P) = 0. \tag{5.16}$$

For this particular permutation all three $\mathcal{M}_q$'s are zero. That is not true for every permutation, as the previous counterexample illustrates. In the coherent basis,

$$P: \quad h_+ \leftrightarrow v_+, \qquad h_- \leftrightarrow v_-. \tag{5.17}$$

The axis changes, but no amplitude is divided between outputs.

5.5 The Grover coin

Define $|s\rangle = (1, 1, 1, 1)^{\mathsf{T}}/2$. The Grover coin is

$$\boxed{G = 2|s\rangle\langle s| - I = \frac{1}{2}\begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}.} \tag{5.18}$$

It is real, symmetric, Hermitian, and unitary, with $G^2 = I$. It has eigenvalue $+1$ on $|s\rangle$ and -1 on its three-dimensional orthogonal complement. Every direction permutation leaves it unchanged. This is a symmetry of the *coin*; only permutations compatible with the shift and geometry can become symmetries of the full walk.

Its probabilities are uniformly $1/4$, giving

$$\mathcal{M}_{\text{split}}(G) = 1, \qquad (\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3})(G) = (1, 1, 1). \tag{5.19}$$

Its coherent action is much more restricted:

$$Gh_+ = v_+, \qquad Gv_+ = h_+, \qquad Gh_- = -h_-, \qquad Gv_- = -v_-. \tag{5.20}$$

In the coherent basis already defined,

$$\widetilde{G} = Q^\dagger G Q = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{5.21}$$

This is the key coin identity behind Grover's exact rectangular modes. Equal output probabilities do not erase the information in phases.

5.6 The unperturbed Fourier coin

5.6.1 Assigning Fourier indices to physical directions

The four-dimensional discrete Fourier matrix has entries $F_{jk} = i^{jk}/2$, $j, k = 0, 1, 2, 3$. It is a unitary, symmetric, fourth root of the identity. A Fourier *coin* additionally requires a choice of which cyclic index is attached to which physical direction. Because the shift is held fixed, different assignments define different walks; a change of numerical basis, by contrast, would transform both S and C and could not change the spectrum.

The *compass assignment* uses the cyclic order (R, U, L, D), so that consecutive indices are consecutive quarter turns. Written in our fixed physical order (R, L, U, D), it is

$$\boxed{F_c = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \\ 1 & -1 & -i & i \end{pmatrix}}. \tag{5.22}$$

This is the unperturbed coin of the project's (now removed) designed Fourier family. The *physical-order assignment* attaches the indices directly to (R, L, U, D):

$$F_p = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}. \tag{5.23}$$

Both are symmetric unitaries with eigenvalues $1, 1, -1, i$ and $F^4 = I$. In the physical square every entry has modulus $1/2$: $\mathcal{M}_{\text{split}} = 1$ and $(\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3}) = (1, 1, 1)$ for both, exactly as for Grover. Neither commutes with J.

5.6.2 The compass coin in the 16-cell map

The coherent square of F_c is a phased permutation of the channels:

$$F_c: \qquad h_+ \mapsto h_+, \qquad v_+ \mapsto h_-, \qquad h_- \mapsto v_+, \qquad v_- \mapsto i\, v_-. \tag{5.24}$$

The exchange $v_+ \leftrightarrow h_-$ changes the wall parity, so two forbidden cells carry weight one and $\mathcal{L}_{\text{SW}}(F_c) = 1/2$. This is a gauge artifact. With

$$\boxed{D = \operatorname{diag}(i, -i, 1, 1), \qquad \widetilde{DF_cD} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \oplus (-1) \oplus (i),} \tag{5.25}$$

all forbidden cells are empty: $\mathcal{L}^*_{\text{SW}}(F_c) = 0$, with $C_+ = i\sigma_x$, $e^{i\alpha} = -1$, $e^{i\beta} = i$ (Fig. 5.1). **Exact result.** The displayed congruence is an identity of 4×4 matrices. The corresponding gauge transformation of the full walk is given in Section 6.7. Up to the global factor i on the symmetric block and the phase i on v_-, the gauge image is Grover's coherent structure.

5.6.3 The physical-order coin

The coherent square of F_p has weight in all sixteen cells (Fig. 5.1), with $\mathcal{L}_{\text{SW}}(F_p) = 9/16$. **Computed for this revision.** A numerical minimization over the gauge phases (169 starting

points) gives $\mathcal{L}^*_{\mathrm{SW}}(F_p) \le 0.073$, and no smaller value was found. F_p is not monomial and has $\mathcal{M}_{q_1} = 1$, so it is in neither the routing family nor the row/column family. It therefore lies outside the four demonstrated reduction families as far as tested. Its full walk was not examined in this revision.

5.7 The Bell coins

Two different matrices have been called *Bell* in the project. They define different walks and are kept apart under the names B_{I} and B_{II}.

5.7.1 The matrix of the earlier report: B_{I}

In this subsection and in Section 6.8, B denotes $B_{\mathrm{I}} = B$. In (R, L, U, D):

$$\boxed{B = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.} \tag{5.26}$$

It is real, symmetric, unitary, and satisfies $B^2 = I$. Each input direction splits into two equal probabilities, giving $\mathcal{M}_{\mathrm{split}} = 1/2$ and $\mathcal{L}_{\mathrm{route}} = 1/2$. Its pair measures are

$$(\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3})(B) = (1, 1, 0), \qquad [B, q_3] = [B, J] = 0. \tag{5.27}$$

It preserves the groups $\{R, D\}$ and $\{L, U\}$, which are not the horizontal and vertical axes.

In the coherent basis its action is

$$B|h_+\rangle = (-|h_+\rangle + |v_+\rangle)/\sqrt{2}, \qquad B|v_+\rangle = (|h_+\rangle + |v_+\rangle)/\sqrt{2}, \tag{5.28}$$

$$B|h_-\rangle = (-|h_-\rangle - |v_-\rangle)/\sqrt{2}, \qquad B|v_-\rangle = (-|h_-\rangle + |v_-\rangle)/\sqrt{2}. \tag{5.29}$$

It preserves the symmetric and antisymmetric coin spaces separately, but it mixes the horizontal and vertical antisymmetric vectors. Their different spatial envelopes will matter in the full walk.

The historical conversion from an earlier report's basis was not independently checked in the supplied session export. It does *not* reproduce the project's Bell IPR figure; the matrix that does is B_{II} below. Its numerical gauge minimum is $\mathcal{L}^*_{\mathrm{SW}}(B_{\mathrm{I}}) \le 1/4$, equal to its fixed-mode value (computed for this revision).

5.7.2 A second matrix called Bell: B_{II}

The project's Bell IPR figure was produced with QWProgramming's Bell coin. In the QWProgramming order (U, D, R, L) its rows are the four Bell states $\Phi^+, \Phi^-, \Psi^+, \Psi^-$ written in the computational basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle \equiv U, D, R, L$. Translated to our physical order (R, L, U, D), it is

$$\boxed{B_{\mathrm{II}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix}.} \tag{5.30}$$

Its columns are exactly the coherent channels:

$$B_{\mathrm{II}}|R\rangle = |h_-\rangle, \quad B_{\mathrm{II}}|L\rangle = |v_-\rangle, \quad B_{\mathrm{II}}|U\rangle = |v_+\rangle, \quad B_{\mathrm{II}}|D\rangle = |h_+\rangle. \tag{5.31}$$

In other words, B_{II} maps the direction basis onto the coherent (Bell-type) basis, which is the origin of its name. It is real orthogonal but *not* symmetric, has eigenphases $\pm\pi/12, \pm 7\pi/12$ and order 24, splits each direction into two equal outputs ($\mathcal{M}_{\mathrm{split}} = 1/2$), and has $(\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3}) = (1, 1, 1)$ and $[B_{\mathrm{II}}, J] \neq 0$.

In the 16-cell map,

$$B_{\mathrm{II}}|h_+\rangle = (|h_-\rangle + |v_-\rangle)/\sqrt{2}, \qquad B_{\mathrm{II}}|v_+\rangle = (|h_+\rangle + |v_+\rangle)/\sqrt{2}, \tag{5.32}$$

$$B_{\mathrm{II}}|h_-\rangle = (|h_-\rangle - |v_-\rangle)/\sqrt{2}, \qquad B_{\mathrm{II}}|v_-\rangle = (-|h_+\rangle + |v_+\rangle)/\sqrt{2}. \tag{5.33}$$

The input h_+ is sent entirely into antisymmetric channels and the input v_- entirely into symmetric ones. Both columns are therefore completely forbidden, and half of the h_- column is forbidden as well. This gives $\mathcal{L}_{\mathrm{SW}}(B_{\mathrm{II}}) = 5/8$. **Computed for this revision.** The numerical gauge minimum is $\mathcal{L}^*_{\mathrm{SW}}(B_{\mathrm{II}}) \leq 1/2$ (169 starting points); no lower value was found.

Exact result. B_{I} and B_{II} are not related by any relabeling of directions, with or without signs. All 24×16 signed permutations Π were checked against both $\Pi B_{\mathrm{II}} \Pi^{\mathsf{T}} = B_{\mathrm{I}}$ and $\Pi B_{\mathrm{II}}^{\mathsf{T}} \Pi^{\mathsf{T}} = B_{\mathrm{I}}$, and none holds. The difference is therefore not a basis-order or sign convention. **Computed for this revision.** On a 20×17 rectangle, B_{II} reproduces the project's Bell IPR figure, while B_{I} gives 37 eigenvalues at each of ± 1 and a scattered IPR pattern instead. The transpose $B_{\mathrm{II}}^{\mathsf{T}}$ gave the same IPR picture as B_{II} at that size. Both Bell walks are kept in these notes under distinct names.

5.8 The modified Grover family

The project constructed a deterministic deformation of that coherent action. Let $c = \cos\theta$, $s = \sin\theta$, and define

$$\boxed{\widetilde{C}_G(\theta) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -c & is \\ 0 & 0 & is & -c \end{pmatrix}, \qquad C_G(\theta) = Q\widetilde{C}_G(\theta)Q^\dagger.} \tag{5.34}$$

The export denotes it $C(\theta)$. We add the subscript G here to distinguish it from the general coin and the Fourier family. In physical direction order,

$$C_G(\theta) = \frac{1}{2}\begin{pmatrix} -c & c & 1+is & 1-is \\ c & -c & 1-is & 1+is \\ 1+is & 1-is & -c & c \\ 1-is & 1+is & c & -c \end{pmatrix}. \tag{5.35}$$

It is unitary and symmetric, commutes with J, and equals Grover at $\theta = 0$. It retains the exchange of h_+, v_+ but couples h_-, v_-. Equivalently,

$$C_G(\theta) = Ge^{-i\theta V}, \qquad V = |h_-\rangle\langle v_-| + |v_-\rangle\langle h_-|. \tag{5.36}$$

The exact direction-pair values are

$$(\mathcal{M}_{q_1}, \mathcal{M}_{q_2}, \mathcal{M}_{q_3}) = (\cos^2\theta, 1, 1). \tag{5.37}$$

At $\theta = \pi/3$, an explicit example is

$$\boxed{C_* = \frac{1}{4}\begin{pmatrix} -1 & 1 & 2+i\sqrt{3} & 2-i\sqrt{3} \\ 1 & -1 & 2-i\sqrt{3} & 2+i\sqrt{3} \\ 2+i\sqrt{3} & 2-i\sqrt{3} & -1 & 1 \\ 2-i\sqrt{3} & 2+i\sqrt{3} & 1 & -1 \end{pmatrix}.} \tag{5.38}$$

The recorded exact coin properties are eigenvalues $1, -1, e^{2\pi i/3}, e^{-2\pi i/3}$, trace and determinant -1, and $C_*^6 = I$. Also $[C_*, G] = 0$. None of these statements alone implies that the full walks commute or have period six: the shift intervenes between coin applications.

5.9 The modified Bell family

5.9.1 Definition: the same rotation as for Grover

The modified Grover coin was obtained as $C_G(\theta) = Ge^{-i\theta V}$, with $V = |h_-\rangle\langle v_-| + |v_-\rangle\langle h_-|$. Applying *the same rotation* to the Bell coin that reproduces the IPR figure defines

$$\boxed{B_{\rm II}(\theta) = B_{\rm II}\, e^{-i\theta V}, \qquad 0 \le \theta \le \pi.} \tag{5.39}$$

Because $B_{\rm II}$ and V are real, $B_{\rm II}(-\theta) = \overline{B_{\rm II}(\theta)}$, so the walk at $-\theta$ is the complex conjugate of the walk at θ, with conjugate spectrum. The interval $[0, \pi]$ is therefore enough. No mirror identity analogous to Grover's $\theta \to \pi - \theta$ relation was found, so the whole interval is used. At $\theta = \pi$, $e^{-i\pi V} = I - 2V^2$ is real and the coin is real again.

5.9.2 What the rotation does in the 16-cell square

Right multiplication by $e^{-i\theta V}$ acts on the input columns h_- and v_- only. With $c = \cos\theta$, $s = \sin\theta$, the coherent square is

$$\widetilde{B}_{\rm II}(\theta) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & is & -c \\ 0 & 1 & -is & c \\ 1 & 0 & c & -is \\ 1 & 0 & -c & is \end{pmatrix}, \qquad \text{columns } (h_+, v_+, h_-, v_-). \tag{5.40}$$

The allowed weight comes from three places. The v_+ column contributes 1. The h_- column contributes $c^2/2$ in its diagonal cell, and the v_- column contributes $s^2/2$ in its diagonal cell. Their sum is $3/2$, independent of θ. Hence

$$\boxed{\mathcal{L}_{\rm SW}\big(B_{\rm II}(\theta)\big) = \frac{5}{8} \quad \text{for every } \theta,} \qquad \text{versus} \qquad \mathcal{L}_{\rm SW}\big(C_G(\theta)\big) = \tfrac{1}{2}\sin^2\theta. \tag{5.41}$$

For Grover the antisymmetric inputs start fully allowed: the allowed weight from those two columns is $2\cos^2\theta$, and the rotation switches leakage on. For $B_{\rm II}$ one antisymmetric column is already fully forbidden. The rotation only moves the surviving allowed half between the two diagonal cells. *The deformation that drives modified Grover out of the standing-wave family therefore does not act on $B_{\rm II}$ through that family at all.*

Computed for this revision. The gauge minimum $\mathcal{L}^*_{\rm SW}(B_{\rm II}(\theta))$ was minimized numerically on 41 values of θ. It equals $1/2$ at $0, \pi/2, \pi$, dips to about 0.444 near $\pi/4$ and $3\pi/4$, is symmetric about $\pi/2$, and never reaches zero (Fig. 7.1). At $\theta = \pi/4$ the coin has $\mathcal{M}_{\rm split} = 0.80$ and $\mathcal{M}_q = (1, 1, 1)$.

Chapter 6

The complete rectangular walk, coin by coin

All complete walks use the same rectangle and boundary rule. We test the candidate structures before interpreting the spectrum. The exactly solved examples come first (permutation, Grover, compass Fourier); the two Bell walks and the deformed families then show what does and does not follow from losing those particular reductions.

6.1 The compatibility test for the complete step

The product identity is

$$[U, Q] = S[\mathcal{C}, Q] + [S, Q]\mathcal{C}. \tag{6.1}$$

For a shift-conserved Q, the second term is zero. Since S is invertible,

$$[U, Q] = 0 \quad \Longleftrightarrow \quad [\mathcal{C}, Q] = 0 \qquad \text{when } [S, Q] = 0. \tag{6.2}$$

This tests a coin against an already specified shift structure. It does not make an arbitrary local coin eigenbasis a conserved mode basis of the walk.

For example, $\widehat{q}_1 = I_{\text{pos}} \otimes q_1$ commutes with S, because wall reversal preserves the axis. The other two physical pair labels are generally changed at the walls. Bell's conserved coin-pair label therefore requires an additional shift check, as the worked example below shows.

6.2 Exact solvability and spectral evidence answer different questions

In this project an explicit geometric reduction is the strongest available evidence for what we call an integrable walk. Examples are a complete set of standing-wave sectors of bounded dimension, independent one-dimensional row/column problems, or deterministically routed basis cycles. The construction must be available without first diagonalizing the whole Floquet matrix. Arbitrary eigenprojectors exist for every finite unitary and would make that criterion empty.

A few conserved parities do not provide such a complete solution. They divide $\mathcal{H}$ into smaller spaces that can still grow with the area and exhibit quantum-chaotic spectral correlations. Conversely, an exact solution need not produce a Poisson-like full spectrum: flat bands, repeated blocks, and arithmetic degeneracies can dominate a naive spacing statistic.

To state the numerical evidence, sort eigenphases within the selected symmetry sector and window, form neighboring gaps s_j, and use

$$\widetilde{r}_j = \frac{\min(s_j, s_{j+1})}{\max(s_j, s_{j+1})}. \tag{6.3}$$

The recorded benchmarks are

$$\langle \widetilde{r} \rangle_{\text{Poisson}} = 2\ln 2 - 1 \simeq 0.3863, \quad \langle \widetilde{r} \rangle_{\beta=1} \simeq 0.5307, \quad \langle \widetilde{r} \rangle_{\beta=2} \simeq 0.5996. \tag{6.4}$$

These local-spacing reference values are shared, in the large-size limit, by the corresponding Gaussian and circular ensembles [2]. A mean near a reference is evidence, not proof of the entire ensemble distribution or its long-range correlations.

Before computing it, one must resolve the surviving unitary symmetries, identify exact degenerate phases, and specify the selected branch or window. Pooling already computed ratio samples is different from merging eigenphases of independent sectors. The latter introduces spacings between levels that cannot repel one another.

6.3 Time reversal and conjugate pairing are different relations

Let K denote complex conjugation in the position–direction basis. For a real walk, $KUK = U$, so a phase $e^{i\varphi}$ has a partner $e^{-i\varphi}$. More generally a commuting antiunitary has that consequence inside the sectors it preserves.

Time reversal of a Floquet step instead obeys

$$AUA^{-1} = U^{-1}, \qquad A \text{ antiunitary.} \tag{6.5}$$

These equations are not interchangeable. The relevant orthogonal-class time reversal has $A^2 = I$. It identifies the appropriate random-matrix benchmark *if* a sector is chaotic; it does not make an exactly solvable walk chaotic.

For any symmetric homogeneous coin, $C = C^{\mathsf{T}}$, the rectangle supplies a direct construction. Let $P_0 = \mathcal{R}\mathcal{J}$, the reflection of both spatial coordinates without changing direction. Then

$$\boxed{A = \mathcal{C}^\dagger P_0 K, \qquad AUA^{-1} = U^{-1}, \qquad A^2 = I.} \tag{6.6}$$

The proof uses $P_0 S P_0 = S^{-1}$, reality of S, and $\bar{C} = C^\dagger$ for a symmetric unitary. All displayed coins are symmetric except B_{II} and the modified Bell family; the detailed sector preserved by an antiunitary must still be checked. Appendix B records the more general reversing criterion already established.

Relation	Type of operation	What it establishes
$[U, Q] = 0$	A linear conserved observable	Its eigenspaces are invariant
$AUA^{-1} = U$	Commuting antiunitary	Conjugate phase partners in sectors preserved by A
$AUA^{-1} = U^{-1}$	Reversing antiunitary	Floquet time reversal; its square must also be checked
$VUV^\dagger = e^{i\alpha}U$	Unitary spectral rotation	Related eigenvalues, generally not conserved eigenspaces of V

These are different uses of the word symmetry. In particular, a spectral pairing or rotation should not automatically be treated as another commuting projector for diagonalization.

6.4 Measuring coupling between the rectangular modes

Let $\mathscr{P} = \{P_a\}$ be a specified complete orthogonal decomposition, with $r_a = \operatorname{Tr} P_a$. Section 4.6 defines these matrices explicitly for the standing-wave family. The block operator $P_b U P_a$ transfers an input component in sector a to output sector b. Define

$$W_{ba} = \|P_b U P_a\|_F^2 , \qquad \sum_b W_{ba} = r_a, \qquad \sum_a W_{ba} = r_b. \tag{6.7}$$

The conditional probability for input P_a/r_a is W_{ba}/r_a. The mean probability of changing the sector, with uniform weight on the full Hilbert space, is

$$\boxed{\mathcal{L}_{\mathscr{P}}(U) = 1 - \frac{1}{D}\sum_a \|P_a U P_a\|_F^2 = \frac{1}{2D}\sum_a \|[U, P_a]\|_F^2 .} \tag{6.8}$$

For a single projector the commutator norm equals $2r_a - 2\,\|P_a U P_a\|_F^2$; summing proves the identity. Therefore

$$\boxed{\mathcal{L}_{\mathscr{P}}(U) = 0 \iff [U, P_a] = 0 \quad \text{for every } a.} \tag{6.9}$$

The existing worst-case test is

$$\eta_{\mathscr{P}}(U) = \max_a \|(I - P_a)U P_a\|_{\text{op}}^2 .$$

The operator norm is the largest output norm for a unit input. It detects complete escape by one input even when the average leakage is small.

For the standing-wave family $[S, P_a] = 0$, so $\mathcal{L}_{\text{SW}}(S\mathcal{C}) = \mathcal{L}_{\text{SW}}(\mathcal{C})$. The coin-only expression still depends on projectors constructed from the rectangle and its reflecting shift.

Writing $\widetilde{C} = Q^\dagger C Q$, the established exact formula is

$$\boxed{\mathcal{L}_{\text{SW}}(U_C) = 1 - \frac{1}{4}\left(\sum_{a,b=1}^{2} |\widetilde{C}_{ab}|^2 + |\widetilde{C}_{33}|^2 + |\widetilde{C}_{44}|^2\right).} \tag{6.10}$$

The symmetric components have a common cosine–cosine profile and can mix within a sector. The horizontal and vertical antisymmetric components have different profiles and must retain their own coin labels to preserve these particular sectors. In particular,

$$\mathcal{L}_{\text{SW}} = 0 \iff \widetilde{C} = C_+ \oplus e^{i\alpha} \oplus e^{i\beta}, \qquad C_+ \in U(2). \tag{6.11}$$

This characterizes one demonstrated solvable family, not every integrable coin.

6.5 Permutation: the full walk has an exact cycle solution

For the displayed permutation coin, both factors of $U_P = S(I \otimes P)$ permute the position–direction basis. Their product is another permutation. Its cycles can differ from the shift's original row and column cycles, but Fourier transformation along each new cycle gives its exact spectrum.

More generally, a monomial coin has

$$C|d\rangle = e^{i\gamma_d}|\pi(d)\rangle. \tag{6.12}$$

The full step has weighted cycles. If one has length L and total accumulated phase Φ, its eigenvalues are

$$\boxed{\lambda_k = e^{i(\Phi + 2\pi k)/L}, \qquad k = 0, \ldots, L-1.} \tag{6.13}$$

This exact routing solution applies even when the original rectangular standing-wave labels are lost. For the displayed P,

$$\mathcal{L}_{\mathrm{SW}}(U_P) = \frac{1}{2}, \qquad \mathcal{M}_{\mathrm{split}}(P) = 0. \tag{6.14}$$

Positive leakage from one mode family is therefore not a proof of nonintegrability. The rotation $\mathcal{R}$ also survives because $[P, J] = 0$, but it is the full cycle construction that solves the walk.

6.6 Grover: strong local splitting, exact collective modes

The coin exchanges h_+, v_+, whose spatial profiles are identical inside $\mathcal{K}_{mn}$, and multiplies each antisymmetric vector by -1. Thus it preserves every projector of $\mathscr{P}_{\mathrm{SW}}$. Since the shift does as well,

$$\boxed{[U_G, P_a] = 0 \quad \text{for every } a, \qquad U_G = \bigoplus_a U_a, \quad \dim U_a \le 4.} \tag{6.15}$$

The local splitting entropy is one while mode leakage is zero:

$$\mathcal{M}_{\mathrm{split}}(G) = 1, \qquad \mathcal{L}_{\mathrm{SW}}(U_G) = 0. \tag{6.16}$$

The same state can therefore spread over physical sites and directions while retaining its collective mode labels.

For $m, n > 0$, in the order $(a_{mn}, b_{mn}, c_{mn}, d_{mn})$, write $c_x = \cos(\pi m/N_x)$, $s_x = \sin(\pi m/N_x)$, and similarly c_y, s_y. The full restricted step is

$$U_{mn} = \begin{pmatrix} 0 & c_x & s_x & 0 \\ c_y & 0 & 0 & s_y \\ 0 & s_x & -c_x & 0 \\ s_y & 0 & 0 & -c_y \end{pmatrix}, \tag{6.17}$$

with characteristic polynomial

$$\det(\lambda I - U_{mn}) = (\lambda^2 - 1)[\lambda^2 + (c_x + c_y)\lambda + 1]. \tag{6.18}$$

With one zero index, the three-dimensional block has the same quadratic times $\lambda - 1$. At $(0, 0)$ the two-dimensional block is the exchange matrix. Terminal singletons have eigenvalue $+1$.

The complete spectrum is consequently

$$n(+1) = N + N_x + N_y, \tag{6.19}$$
$$n(-1) = N - N_x - N_y + 2, \tag{6.20}$$

and, for every $(m, n) \neq (0, 0)$ with $0 \le m < N_x$, $0 \le n < N_y$, one pair $e^{\pm i\varphi_{mn}}$ satisfying

$$\boxed{\cos\varphi_{mn} = -\frac{1}{2}\left(\cos\frac{\pi m}{N_x} + \cos\frac{\pi n}{N_y}\right).} \tag{6.21}$$

Coincident values from different pairs have their multiplicities added. The two flat-band multiplicities sum to $2N + 2$. About half the full spectrum is therefore flat; a full-spectrum ratio mean is not a generic Poisson sample.

All rectangular geometric reflections survive Grover. A standing-wave sector has parities $p_x = (-1)^m$, $p_y = (-1)^n$. A square has additional D_4 structure, which must be resolved when analyzing its dispersive phases. The exact solution, rather than a particular spacing mean, establishes the rectangular reduction.

Recorded numerical result. The source containing general Grover billiards verified the rectangular analytic spectra against direct sector calculations, with maximum errors 8.4×10^{-15} at 12×8 and 3.2×10^{-14} at 41×25. Its resolved dispersive means vary with arithmetic degeneracies: for instance the recorded whole-branch pooled values are 0.3895 on 40×24 and 0.3733 on 41×25. Such variation does not invalidate the exact mode solution.

Computed for this revision. *Eigenstates and sectors on* 40×35. Full diagonalization of U_G ($V = 1400$, $\dim U = 5600$) gives $n(+1) = 1475 = N + N_x + N_y$ and $n(-1) = 1327 = N - N_x - N_y + 2$. Resolving the four parity sectors with the full operators $\mathcal{P}_x, \mathcal{P}_y$ gives:

Sector (p_x, p_y)	Rank	Flat $+1$	Flat -1	Phases in $(0, \pi)$	Zero gaps	$\langle r \rangle$
$(+,+)$	1420	378	324	359	4	0.384
$(+,-)$	1380	377	323	340	2	0.383
$(-,+)$	1420	360	340	360	0	0.385
$(-,-)$	1380	360	340	340	0	0.369

In every sector the phases agree with Eq. (6.21) to 5.6×10^{-15}, and $\langle r \rangle$ is close to the Poisson value. The spatial inverse participation ratio $\mathrm{IPR} = \sum_{\mathbf{r}} P(\mathbf{r})^2$, with $P(\mathbf{r}) = \sum_d |\psi_d(\mathbf{r})|^2$, is shown in Fig. 6.1 (right) and resolved by sector in Fig. 6.2. Two features are explained by the exact solution:

1. the dispersive states organize into smooth families, because each eigenstate is a product standing wave from a block of dimension at most four;
2. the vertical spikes at 0 and $\pm\pi$ come from the flat bands. Inside a degenerate eigenspace the eigensolver returns an arbitrary basis, so these IPR values (up to 0.82) are basis-dependent and carry no physical meaning.

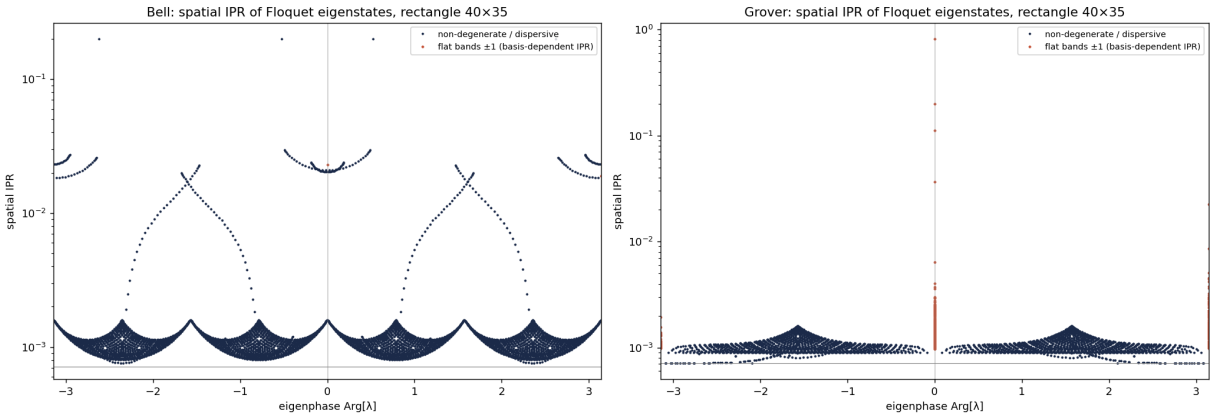


Figure 6.1: Spatial IPR of the Floquet eigenstates on a 40×35 rectangle. Left: Bell B_{II}. Right: Grover, where the eigenvalues at exactly ± 1 are drawn in red because their IPR is basis-dependent. The grey line is $1/V$. This reproduces the project's two IPR figures (the Bell one with B_{II}, not B_{I}).

6.7 Compass Fourier: an exact solution hidden by a gauge

Write $U_F = S[I_{\mathrm{pos}} \otimes F_c]$. The gauge identity of Appendix A with parameters $\alpha = \pi$, $\beta = 0$ reads

$$\boxed{W|x, y, d\rangle = (-1)^x \ \mathrm{diag}(-i, i, 1, 1)_{dd} |x, y, d\rangle, \qquad W U_F W^\dagger = S\,[I_{\mathrm{pos}} \otimes D F_c D],} \tag{6.22}$$

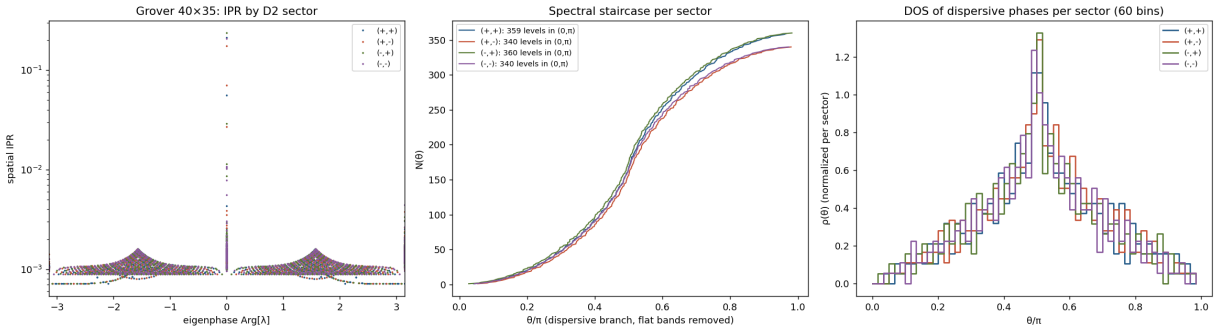


Figure 6.2: Grover on 40×35, resolved into the four (p_x, p_y) sectors: IPR, spectral staircase of the dispersive branch, and its density of states. The density peaks at $\varphi = \pi/2$, the van Hove point of the square lattice.

with $D = \mathrm{diag}(i, -i, 1, 1)$ as in Eq. (5.25). The transformed coin is in the solvable family with $C_+ = i\sigma_x$, $e^{i\alpha} = -1$, $e^{i\beta} = i$. The walk is therefore exactly solved by the gauge-transformed standing-wave projectors $W^\dagger P_a W$, with blocks given by Eq. (4.23):

- the horizontal terminal states carry eigenvalue $+1$ (N_y of them);
- the vertical terminal states carry eigenvalue $-i$ (N_x of them);
- the $(0, 0)$ block has eigenvalues $\pm i$.

Computed for this revision. The operator identity holds to 5×10^{-32}, and the union of block spectra matches direct diagonalization of U_F to 5.7×10^{-15} (7×5) and 1.3×10^{-14} (8×6).

This replaces the earlier record that the Fourier endpoint was only *numerically* identified with the gauge orbit, without an explicit transformation. A second check, also computed for this revision, on 7×5 shows that U_F commutes with $\mathcal{P}_y$ and with the modified rotation $\mathcal{R}\widehat{\Gamma}$, $\widehat{\Gamma} = I \otimes \mathrm{diag}(1, 1, -1, -1)$, but not with $\mathcal{R}$ or $\mathcal{P}_x$. The spectral statistics of this exactly solved walk were not computed in this revision.

The physical-order coin F_p commutes with none of $\mathcal{R}, \mathcal{R}\widehat{\Gamma}, \mathcal{P}_x, \mathcal{P}_y$ on 7×5. Its full-walk status is open (Section 5.6).

6.8 Bell B_{I}: some symmetries survive, but a full solution is not established

Here $B = B_{\mathrm{I}}$ and $U_B = S[I_{\mathrm{pos}} \otimes B_{\mathrm{I}}]$.

For the explicit Bell matrix, $[B, J] = 0$, so $\mathcal{R}$ commutes with U_B. The walk is real, hence conjugate-paired, and it has time reversal as described above. The separate rectangular mirror symmetries do not follow from this coin.

The coin preserves the q_3 pair label, but a boundary reversal exchanges those pairs. Thus $[B, q_3] = 0$ is not a conserved full-walk label. The established mean leakage of that two-sector label is

$$\mathcal{L}_{q_3}(U_B) = \frac{N_x + N_y}{2N_xN_y}. \tag{6.23}$$

There are $2N_x + 2N_y$ blocked directed channels among $4N$ basis states. An appropriate wall input can cross the pair partition with probability one, so the worst-case leakage is one even when the mean is small.

The Bell action on an antisymmetric horizontal mode contains

$$B|g_m^x f_n^y; h_-\rangle = -\frac{1}{\sqrt{2}}\left(|g_m^x f_n^y; h_-\rangle + |g_m^x f_n^y; v_-\rangle\right). \tag{6.24}$$

The second term has the wrong envelope for the vertical antisymmetric vector in $\mathcal{K}_{mn}$, which requires $f_m^x g_n^y$. For the same nonterminal index, $\langle f_m^x, g_m^x\rangle = 0$. Half the antisymmetric weight leaks while the symmetric states remain within their sectors, giving

$$\boxed{\mathcal{L}_{\mathrm{SW}}(U_B) = \frac{1}{4}.} \tag{6.25}$$

Allowing deterministic relabeling of those mode sectors does not reduce this value. Basis-state routing leakage is 1/2.

Recorded numerical result. On 25×19, the two rotation sectors give the half-circle mean 0.4872 each, with 22 eigenvalues at each of ± 1 in each sector. On 29×22, the corresponding means are 0.4940 and 0.4697.

Unresolved. These results neither construct an integrable Bell reduction nor demonstrate full random-matrix universality. Earlier broad claims that the displayed Bell coin was already proved integrable are not retained. Its complete structure remains an open part of the project.

6.9 Bell B_{II}: a regular spectrum outside the tested families

Write $U_{B_{\mathrm{II}}} = S[I_{\mathrm{pos}} \otimes B_{\mathrm{II}}]$.

Symmetries. $[B_{\mathrm{II}}, \sigma_x]$, $[B_{\mathrm{II}}, \sigma_y]$ and $[B_{\mathrm{II}}, J]$ are all nonzero, so by Eq. (4.24) none of $\mathcal{P}_x, \mathcal{P}_y, \mathcal{R}$ survives. **Computed for this revision.** Searches were run on 7×5 and 8×6, with 7×6 and 8×5 for the unitary case. They covered operators of the form (a rectangle site map in D_2) $\times$ (a constant or sublattice-dependent 4×4 coin matrix). Within this family they found:

- no commuting unitary other than the identity;
- no reversing operator $PUP^{-1} = U^{\mathsf{T}}$;
- no spectral rotation $VUV^{\dagger} = cU$ with $c \in \{-1, \pm i\}$.

These restricted searches are not absence theorems. Since B_{II} is not symmetric, the time-reversal construction of Section 6.3 does not apply either. The walk is real, so $KUK = U$ pairs $e^{i\varphi}$ with $e^{-i\varphi}$, and only $(0, \pi)$ is used.

Spectrum. Computed for this revision. On 40×35 (Figs. 6.1 left and 6.3):

- one eigenvalue at each of ± 1, 2799 in $(0, \pi)$, and exact conjugate pairing;
- $\langle r\rangle = 0.392$ on the half circle, with 14 zero gaps;
- 5304 of the 5600 eigenvalues belong to exact $\pi/2$ quartets: $e^{i\varphi}$, $ie^{i\varphi}$, $-e^{i\varphi}$ and $-ie^{i\varphi}$ all appear to 10^{-9}. This is why the bulk IPR pattern repeats with period $\pi/2$ and the density of states peaks at $\pm\pi/4, \pm 3\pi/4$. No operator was found that implements the quartets.

The 296 levels outside the quartets form the arcs near 0 and $\pm\pi$ and the branches rising toward $\pm\pi/2$. They also include four corner states.

Corner states. Computed for this revision. Four states sit at $\varphi = \pm\pi/6, \pm 5\pi/6$, exact to 10^{-12}. They are exponentially localized at the corners $(N_x, 1)$ and $(1, N_y)$ and have IPR 0.200 on every size tested: 16×13, 16×14, 17×13, 17×14, 20×17 and 40×35. On commensurate rectangles such as 18×12, 24×12 and 16×16, a bulk level coincides exactly with a corner state.

The eigensolver then returns an arbitrary mixture, and the plotted IPR becomes basis-dependent. This explains the values near 0.18 and 0.012 at these phases in the project's original Bell figure.

Placement among the reduction families. The forbidden weight is 5/8 and the numerical gauge minimum 1/2. The coin is not monomial ($\mathcal{M}_{\text{split}} = 1/2$), and it fully mixes the axis label ($\mathcal{M}_{q_1} = 1$), so it is not of row/column type. **Unresolved.** The nearly Poisson half-circle statistic and the organized IPR families point to an exact structure. It is not supplied by any of the four demonstrated reduction families, and its origin, including the $\pi/2$ quartets, is an open question.

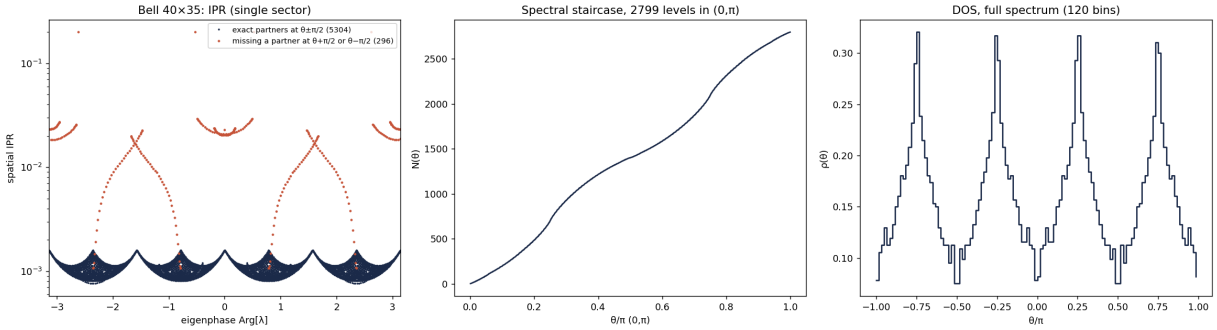


Figure 6.3: Bell B_{II} on 40×35. Left: IPR, with levels outside the exact $\pi/2$ quartets in red. Centre: spectral staircase on $(0, \pi)$. Right: density of states of the full circle.

6.10 Modified Grover: how the coin couples mode labels

Write $U_G(\theta) = S[I_{\text{pos}} \otimes C_G(\theta)]$; thus $U_G(0) = U_G$.

The odd components that $C_G(\theta)$ couples require different envelopes: $g_m^x f_n^y$ for h_- and $f_m^x g_n^y$ for v_-. The onsite coin changes the coin vector without changing the spatial profile. Its output must therefore be re-expanded in other mode labels. The cross-mode coefficients contain

$$\langle f_p^x, g_m^x \rangle \langle g_q^y, f_n^y \rangle. \tag{6.26}$$

Reflection parity makes a nonzero term require both $p + m$ and $q + n$ odd. Many fine labels can couple while their combined rotation parity remains fixed.

The established exact diagnostics are

$$\boxed{\mathcal{L}_{\text{SW}}(U_G(\theta)) = \tfrac{1}{2}\sin^2\theta, \qquad \mathcal{M}_{\text{dir}}(C_G(\theta)) = \cos^2\theta.} \tag{6.27}$$

Thus decreasing the minimum direction-pair score can increase this spatial-mode leakage. These numbers measure different structures. At $\theta = \pi/3$ they are 3/8 and 1/4, respectively. At $\theta = 0$ and π, the fixed standing-wave blocks are preserved again.

6.10.1 The surviving symmetries

The rotation $\mathcal{R}$ survives throughout. The verified time reversal $A = \mathcal{C}_G(\theta)^\dagger \mathcal{J} K$ squares to one and preserves its sectors. The commuting antiunitaries $\mathcal{P}_x K$ and $\mathcal{P}_y K$ also preserve each rotation sector, pairing φ with $-\varphi$. Here $\mathcal{C}_G(\theta) = I_{\text{pos}} \otimes C_G(\theta)$.

Define $\Gamma = \text{diag}(1, 1, -1, -1)$ and $\widehat{\Gamma} = I_{\text{pos}} \otimes \Gamma$. Another exact identity is

$$U_G(\pi - \theta) = -\widehat{\Gamma} U_G(\theta) \widehat{\Gamma}. \tag{6.28}$$

At $\theta = \pi/2$, $\widehat{\Gamma} U \widehat{\Gamma} = -U$, and the additional recorded quarter-circle relation is

$$W_0|x, y, d\rangle = (-1)^{x+y} \operatorname{diag}(-i, i, -i, i)_{dd}|x, y, d\rangle, \tag{6.29}$$

$$V_0 = W_0 \mathcal{P}_x, \qquad V_0 U V_0^\dagger = iU, \tag{6.30}$$

$$V_0 \mathcal{R} = (-1)^{N_x+N_y+1} \mathcal{R} V_0. \tag{6.31}$$

An operator rotating the spectrum by a phase is not a commuting symmetry. These extra relations make this parameter special.

6.10.2 Recorded spectral evidence

For generic parameters the audit selects $10^{-7} < \varphi < \pi - 10^{-7}$ in each rotation sector. It uses noncyclic gaps, sets gaps below 10^{-9} to zero, keeps isolated zero-gap ratios, and omits undefined 0/0 ratios with a count. At $\theta = \pi/3$, the recorded and independently reproduced values are

Rectangle	$\langle r \rangle_+$	Ratios	$\langle r \rangle_-$	Ratios
17×13	0.534116	218	0.539212	219
23×17	0.523110	388	0.526623	389
31×23	0.536581	710	0.532719	711
41×29	0.522816	1186	0.519234	1187

The source reports aspect-ratio and distribution checks as well; Appendix C preserves their scope. These results support orthogonal-class local repulsion at a deterministic coin. They do not establish an exhaustive absence of other conserved structures.

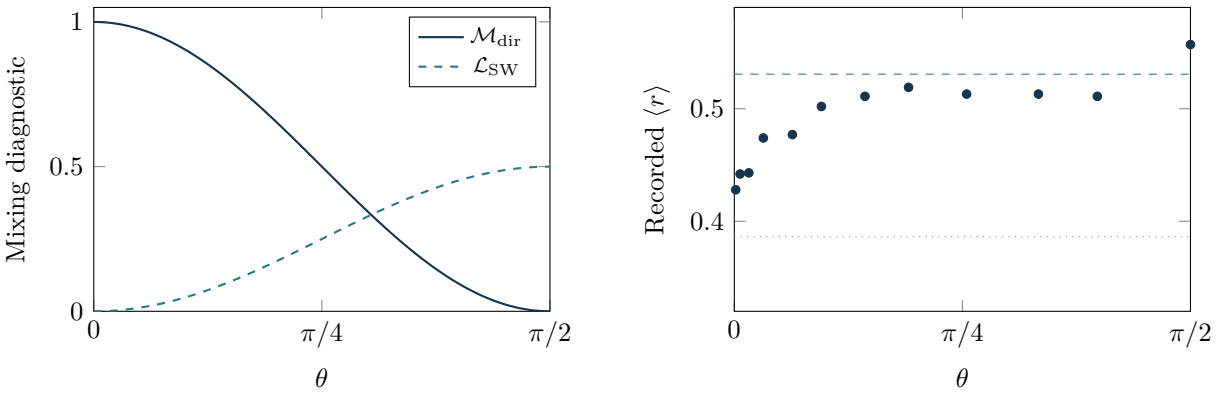


Figure 6.4: Left: the already established exact diagnostics for the modified Grover family. Right: recorded half-circle, rotation-resolved pooled means on 25×19, redrawn from the supplied export; no spectra were recomputed. Dashed and dotted references are the orthogonal and Poisson values. The degenerate Grover endpoint is not plotted. The point $\pi/2$ has additional exact symmetries and cannot be read as a generic crossover endpoint.

6.10.3 What zero direction-pair mixing means at the special point

At $\theta = \pi/2$,

$$C_G(\pi/2) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1+i & 1-i \\ 0 & 0 & 1-i & 1+i \\ 1+i & 1-i & 0 & 0 \\ 1-i & 1+i & 0 & 0 \end{pmatrix}. \tag{6.32}$$

It always swaps the axes, so $\mathcal{M}_{q_1} = 0$, while a definite input direction still splits into two outputs. Its splitting entropy is $1/2$, and $\mathcal{L}_{\mathrm{SW}} = 1/2$.

Recorded half-circle means are about 0.55 on tested odd-by-odd rectangles, but only $0.4361, 0.4487$ in the two sectors on 29×18. The latter half-circle contains degenerate phases at i; its quoted filtering discards undefined pairs. The supplied teaching source additionally records quarter-window checks giving $0.5554, 0.5495, 0.5565$ on $17 \times 13, 23 \times 17, 25 \times 19$, respectively, and approximately $0.4386, 0.4504$ in the two sectors on 29×18. Symmetry-related copies are not independent samples. The parity effect remains unexplained; the point is not an established all-size chaos example.

6.10.4 Results computed for this revision: the full interval on 31×23

Computed for this revision. The rotation sectors are resolved with the full operator $\mathcal{R}$ (site rotation $\times$ J). Within each sector only the half circle $0 < \varphi < \pi$ is used, because $\mathcal{P}_x K$ pairs φ with $-\varphi$; flat ± 1 phases are removed. At $\theta = 0$ the four parity sectors are resolved instead. We also re-verified that

- $\mathcal{R}$ is the only commuting operator among (site map) $\times$ (constant coin matrix);
- a time reversal with square $+1$ exists, and $\mathcal{P}_x K$, $\mathcal{P}_y K$ commute with the walk;
- $U_G(\pi - \theta) = -\widehat{\Gamma} U_G(\theta) \widehat{\Gamma}$ holds to 4×10^{-15}.

The finite-size orthogonal reference was obtained from GOE matrices of the same two sector sizes, processed identically: 0.530 ± 0.010 (20 replicas).

θ/π	0	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
$\langle r \rangle$	0.411	0.476	0.488	0.500	0.512	0.521	0.521	0.535	0.537	0.509	0.548

Statistical errors are about ± 0.007; at $\theta = 0$ the error is ± 0.011 with 704 ratios, and for $\theta > 0$ there are 1421 ratios. Additional angles give 0.486 ($\pi/32$), 0.502 ($\pi/16$) and 0.535 ($\pi/3$). The last agrees with the recorded rotation-sector values 0.5366 and 0.5327 on the same rectangle in the table above, which is an independent reproduction.

Size checks at $\theta = \pi/4$ give 0.498 ± 0.011 (21×15) and 0.515 ± 0.005 (41×31). At $3\pi/8$ they give 0.512 ± 0.010 and 0.516 ± 0.005. The 41×31 values lie slightly below the orthogonal reference. The point 0.45π is about 2σ below its neighbours and was not rechecked.

The IPR collage, Fig. 6.5, shows the same crossover in the eigenstates. At $\theta = 0$ there are smooth families and red flat bands. By $\theta = \pi/32$ the flat bands are gone and the families begin to break. From about $\pi/4$ the bulk IPR collapses into a narrow, featureless band, as expected for eigenstates spread over the whole rectangle.

6.11 Modified Bell: the exact structures break, chaos is not reached

Write $U_B(\theta) = S[I_{\mathrm{pos}} \otimes B_{\mathrm{II}}(\theta)]$.

Symmetries. Computed for this revision. For $\theta \neq 0, \pi$, the search described for B_{II} was extended to commuting antiunitaries $RU^* = UR$ and to reversing antiunitaries $RU^* = U^\dagger R$, with the same restricted family of R on 7×5. It found nothing besides the identity. The coin is complex and not symmetric, so neither the reality pairing nor the Section 6.3 time reversal is available. A single sector and the *full* circle are used, with cyclic gaps. If no antiunitary exists, the relevant

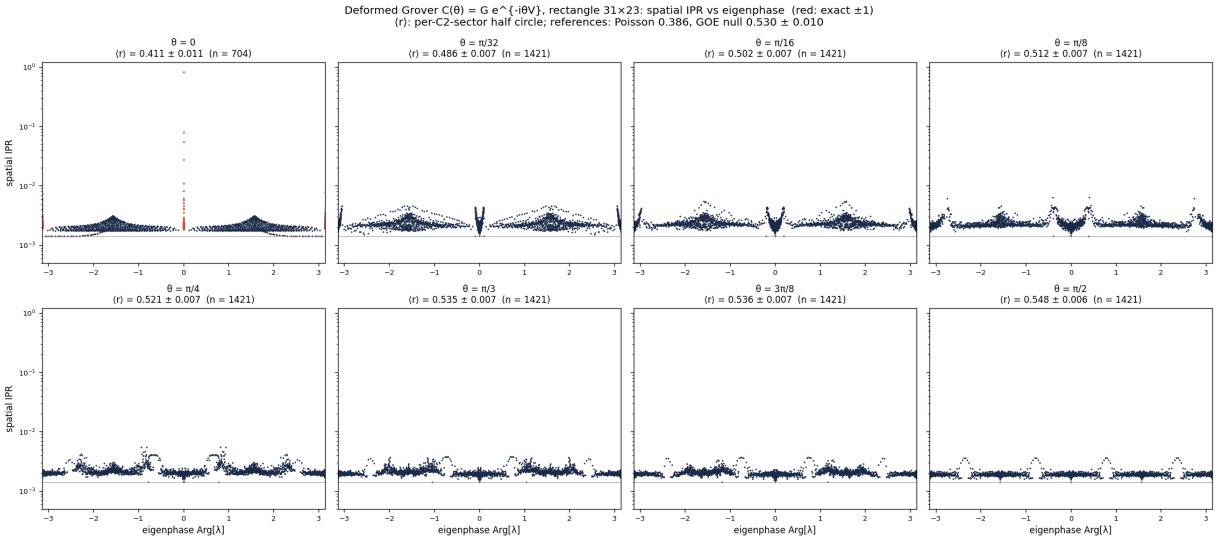


Figure 6.5: Spatial IPR of the modified Grover walk $U_G(\theta)$ on 31×23 for eight values of θ. Each panel title gives the sector-resolved half-circle $\langle r \rangle$. Red points are exact ± 1 eigenvalues, with basis-dependent IPR.

reference would be the unitary class, $\langle r \rangle \approx 0.600$. The finite-size CUE value for 2852 levels was 0.599. At $\theta = 0, \pi$ the walk is real and the half circle is used.

Exact structures disappear. Computed for this revision. On 31×23, the fraction of eigenvalues in exact $\pi/2$ quartets is 0.926 at $\theta = 0$ and at $\theta = \pi$. It is exactly 0 at the nine interior grid points $\theta = 0.1\pi, \ldots, 0.9\pi$. The conjugate pairing behaves the same way: 100% at the endpoints and 0% in between.

Level statistics. Computed for this revision. On 31×23 (half circle at the endpoints with 1424 ratios, full circle otherwise with 2852 ratios, errors about ± 0.005):

θ/π	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$\langle r \rangle$	0.402	0.431	0.453	0.470	0.495	0.457	0.495	0.484	0.467	0.445	0.399

Additional angles give 0.409 ($\pi/16$), 0.440 ($\pi/8$), 0.483 ($\pi/4$), 0.502 ($5\pi/8$) and 0.496 ($3\pi/4$). At $\theta = 0.4\pi$ the size dependence is 0.500 ± 0.007 (21×15), 0.495 ± 0.005 (31×23) and 0.488 ± 0.004 (41×31), drifting slightly downward. At $\pi/2$ it is 0.479 ± 0.008 (21×15) and 0.457 ± 0.005 (31×23). The statistic stays well below both the unitary and the orthogonal references over the whole interval.

Eigenstates. In the IPR collage, Fig. 6.6, the bulk keeps its regular scalloped pattern at every angle. Only the states near the walls, the arcs and isolated points with IPR between about 0.01 and 0.5, are scrambled. At $\theta = \pi$ the full regular pattern returns.

An explanation that was tested and rejected. One might think the rotation leaves the bulk band structure unchanged and acts only through the walls. **Computed for this revision.** This is false. On the infinite lattice, for 200 random wavevectors, the bulk eigenphases of $\Lambda(k_x, k_y) B_{\mathrm{II}}(\theta)$ differ from those at $\theta = 0$ by up to 0.21, 0.61, 0.97 and 1.24 rad at $\theta = 0.3, 0.9, 1.5, 2.5$. Why the bulk eigenstates stay organized remains unexplained.

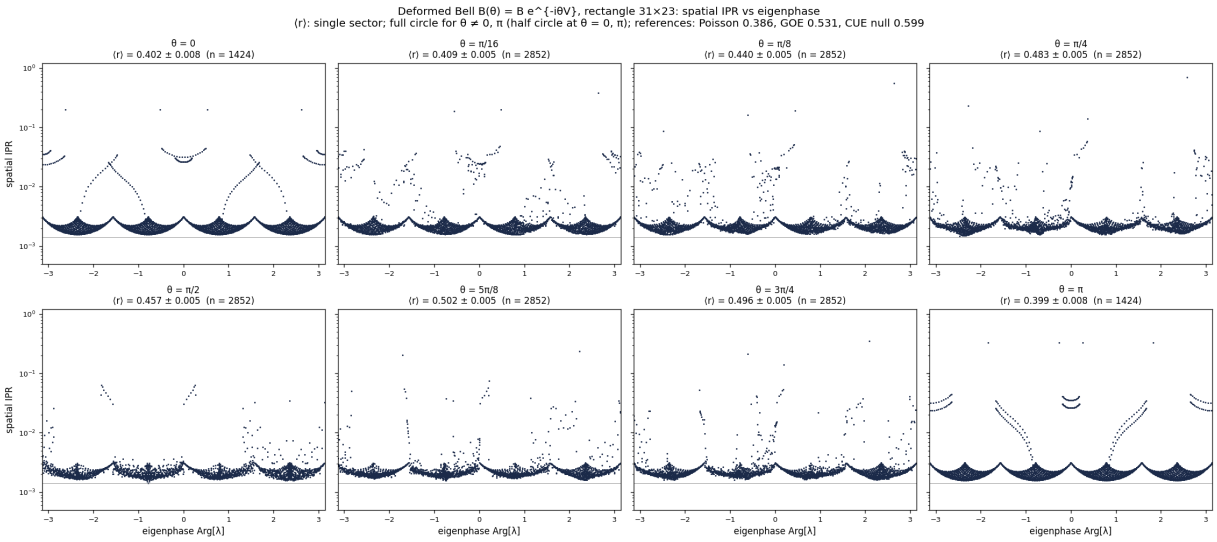


Figure 6.6: Spatial IPR of the modified Bell walk $U_B(\theta)$ on 31×23 for eight values of $\theta \in [0, \pi]$. Each panel title gives $\langle r \rangle$: single sector, full circle for $\theta \neq 0, \pi$.

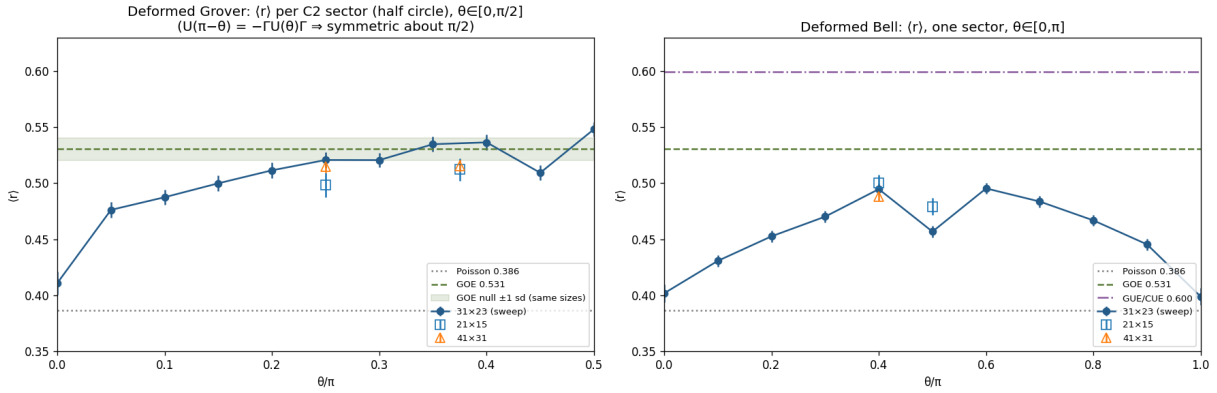


Figure 6.7: $\langle r \rangle$ versus θ for the modified Grover (left; rotation sectors, half circle; band = finite-size GOE null ± 1 s.d.) and modified Bell (right; single sector) walks on 31×23, with size checks on 21×15 and 41×31.

6.12 What the examples have established

Coin	Established full-walk structure	Current conclusion
Identity I	The shift's own row and column cycles (Section 4.7)	Exact; maximally degenerate spectrum
Permutation P	Complete deterministic basis cycles	Exact cycle solution
Grover G	Complete standing-wave blocks, size at most four	Exact rectangular solution; Poisson-like resolved statistics, flat bands
Compass Fourier F_c	Gauge-transformed standing-wave blocks (explicit W)	Exact rectangular solution; statistics not computed
Physical Fourier F_p	None found; outside tested families	Open; walk not examined
Bell B_{I}	Rotation and antiunitary relations; $\mathcal{L}_{\mathrm{SW}} = \mathcal{L}^*_{\mathrm{SW}} = \frac{1}{4}$	Intermediate recorded $\langle r \rangle$; classification open
Bell B_{II}	Reality pairing; exact $\pi/2$ quartets for 95% of levels; corner states; no geometric symmetry found	Regular-looking ($\langle r \rangle \approx 0.39$) but outside tested families; open
Modified Grover	Rotation and time reversal; envelope leakage $\frac{1}{2}\sin^2\theta$	Orthogonal-class-like local statistics from $\theta \approx \pi/4$ at tested sizes
Modified Bell	No symmetry found for $\theta \neq 0, \pi$; quartets and pairing destroyed; $\mathcal{L}_{\mathrm{SW}} = 5/8$ constant	Exact structures broken, statistics intermediate (≤ 0.50); no RMT class reached

6.13 A checklist for teaching a selected coin

1. Read the explicit matrix in the fixed direction order. Its name is not a shift convention or a symmetry proof.
2. Return to the row/column, parity and standing-wave structures supplied before choosing the coin.
3. Test the full-space commutators. Distinguish commuting symmetries from reversers, conjugate pairing and spectral rotations.
4. Display a complete cycle or mode construction when one exists. A few surviving parity sectors only organize the unresolved dynamics.
5. State spectral evidence with its sectors, windows, degeneracy rule and finite-size scope. For both Bell walks, keep the classification open.
6. Draw the 16-cell coherent square with the envelope mask, and compute its gauge minimum, before interpreting any leakage.

The full reasoning chain

First identify the structures supplied by the shift and rectangle. Next test how the coin acts on them. A complete explicit reduction can establish exact solvability; a surviving coarse symmetry only defines sectors. If no complete reduction is known, use correctly resolved spectral evidence and state its scope. Neither local direction splitting nor leakage from one selected mode family is, by itself, a necessary-and-sufficient detector of quantum chaos.

Chapter 7

Routes out of exact solvability in the fixed rectangle

The rectangle and the stay-and-reverse shift are fixed; only the coin changes. In this chapter, *breaking integrability* is used in a deliberately operational sense, with two parts. Structurally, the walk leaves every demonstrated reduction family. Spectrally, correctly resolved local statistics move from Poisson-like values toward a random-matrix reference. Neither part is decisive on its own: a structure outside the tested families may exist, and a spacing statistic is not a proof. The goal here is to describe the routes observed so far, not to decide whether they are one mechanism.

7.1 The reduction families as label algebras supplied by the geometry

Each demonstrated family is a set of projectors built from the rectangle and the shift *before* the coin is chosen, together with the coin condition that keeps them conserved:

Family	Geometric projectors	Coin condition	Fingerprint in the 16-cell map
Rows/columns	P_y^H, P_x^V	$C = C_H \oplus C_V$ in $\{R, L\} \mid \{U, D\}$	Physical square block diagonal by axis
Standing waves	$P_a \in \mathscr{P}_{\mathrm{SW}}$	$\widetilde{C} = C_+ \oplus e^{i\alpha} \oplus e^{i\beta}$	No weight in the ten forbidden cells
Gauge images	$W^\dagger P_a W$	$\mathcal{L}^*_{\mathrm{SW}} = 0$	Forbidden cells empty after $C \to DCD$
Routing	Position–direction basis	Monomial coin	One nonzero cell per physical column

In the language of Appendix B, each family is a commutative label algebra. Exact solvability within the family means that conjugation by U maps the algebra into itself, so that its escape $\mathcal{M}^{\mathrm{alg}}$ vanishes.

7.2 Where each coin sits

Coin	$\mathcal{L}_{\mathrm{SW}}$	$\mathcal{L}^*_{\mathrm{SW}}$	$\mathcal{M}_{\mathrm{split}}$	$\mathcal{M}_{q_1}$	$[C,J]=0$	$C=C^{\mathsf{T}}$	Family / status
I	0	0	0	0	yes	yes	All four families; exact, maximally degenerate
P	1/2	$\leq 1/2$	0	0	yes	yes	Routing; exact
G	0	0	1	1	yes	yes	Standing waves; exact
F_c	1/2	0	1	1	no	yes	Gauge image; exact
F_p	9/16	≤ 0.073	1	1	no	yes	None found; open
B_{I}	1/4	$\leq 1/4$	1/2	1	yes	yes	None found; open
B_{II}	5/8	$\leq 1/2$	1/2	1	no	no	None found; regular-looking, open
$C_G(\theta)$	$\frac{1}{2}\sin^2\theta$	$\frac{1}{2}\sin^2\theta$	varies	$\cos^2\theta$	yes	yes	Leaves standing waves for $0<\theta<\pi$
$B_{\mathrm{II}}(\theta)$	5/8	0.444–0.5	varies	1	no	no	None found; exact structures broken

Values marked $\leq$ are numerical gauge minima, which are upper bounds. The entries for $C_G(\theta)$ are exact; Appendix A gives the proof of its gauge minimum. The splitting entropy of C_G is 0.906 at $\pi/4$ and 0.772 at $\pi/3$; that of $B_{\mathrm{II}}(\pi/4)$ is 0.80.

7.3 Route 1: envelope mismatch (modified Grover)

The rotation $e^{-i\theta V}$ acts on the antisymmetric inputs, which for Grover start entirely in their allowed diagonal cells. It moves the weight $\sin^2\theta$ of each into the cell $h_- \leftrightarrow v_-$. That cell is forbidden because it exchanges the envelope $g^x f^y$ with $f^x g^y$ (Section 4.10). Physically, a standing wave whose sine runs along x is converted, at every site, into the channel whose sine must run along y. The output has to be re-expanded in other mode labels, with overlaps $\langle f^x_p, g^x_m\rangle\langle g^y_q, f^y_n\rangle$ that couple many (p,q) to (m,n) ($p+m$ and $q+n$ odd). On the recorded 9×7 example this coupling already merges the 79 standing-wave sectors into three components (Appendix B).

Three findings show that this route is genuine and not a basis artifact:

1. the leakage cannot be removed by a gauge: $\mathcal{L}^*_{\mathrm{SW}} = \mathcal{L}_{\mathrm{SW}} = \frac{1}{2}\sin^2\theta$, exactly;
2. $\langle r\rangle$ rises together with the leakage, from 0.41 to the orthogonal reference by $\theta \approx \pi/4$ (Fig. 7.1(a));
3. the eigenstates spread over the whole rectangle (Fig. 6.5).

The rise of $\langle r\rangle$ is steep at small θ, where the leakage is still small: no universal leakage threshold exists (Appendix A).

7.4 Route 2: leaving deterministic routing

Recorded numerical result. The near-monomial family $e^{-i\epsilon H/2} P e^{-i\epsilon H/2}$ of Appendix C starts from an exact routing solution. On 25×19 it gave $\langle r\rangle = 0.18$, 0.35 and 0.517 at $\epsilon = 0.02$, 0.2 and 0.8. The degenerate routing spectrum is lifted first, and orthogonal-class-like values appear only at strong deformation. This route breaks a different label algebra, the position–direction basis.

7.5 Route 3: coupling independent rows and columns

Recorded numerical result. The family $C_{\mathrm{rot}}(\varphi) = B_\eta e^{-i\varphi P}$ starts from independent rows and columns. Its exact row/column leakage is $\sin^2\varphi$, yet its recorded statistics stay low: the aggregate over $\varphi \in [0.6, 1.3]$ is 0.416, against 0.526 for modified Grover on the same grid. Leaving one family with large leakage therefore did not, by itself, produce random-matrix statistics. This is a direct counterexample to reading any single leakage as a chaos measure.

7.6 Bell: regular outside the families, and a deformation that breaks without chaos

B_{II} already violates the envelope rule strongly ($\mathcal{L}_{\mathrm{SW}} = 5/8$, $\mathcal{L}^*_{\mathrm{SW}} \le 1/2$) and is neither monomial nor axis-preserving. Nevertheless its walk shows nearly Poisson half-circle statistics, organized IPR families, exact $\pi/2$ quartets for 95% of the levels, and exact corner states. Whatever keeps it regular is *not* one of the four label algebras.

Applying the Grover rotation to it gives the following:

- the forbidden weight is invariant, $\mathcal{L}_{\mathrm{SW}} = 5/8$ exactly, and the gauge minimum moves only between 0.444 and 0.5;
- every exact spectral structure known for the undeformed walk disappears for $0 < \theta < \pi$: the quartets and the conjugate pairing go to 0%;
- $\langle r \rangle$ never exceeds about 0.50, and at $\theta = 0.4\pi$ it decreases slightly with size;
- the bulk eigenstates remain organized (Figs. 6.6 and 7.1(b)).

The rotation was designed to destroy Grover's particular solvable structure, the envelope compatibility of h_- and v_-. For B_{II} that structure was never present, so "the same logic" is not the same mechanism: the deformation breaks B_{II}'s exact structures without reaching a random-matrix regime at the tested sizes. **Recorded numerical result.** B_{I}, with $\mathcal{L}^*_{\mathrm{SW}} \le 1/4$, likewise has intermediate rotation-sector values 0.47–0.49.

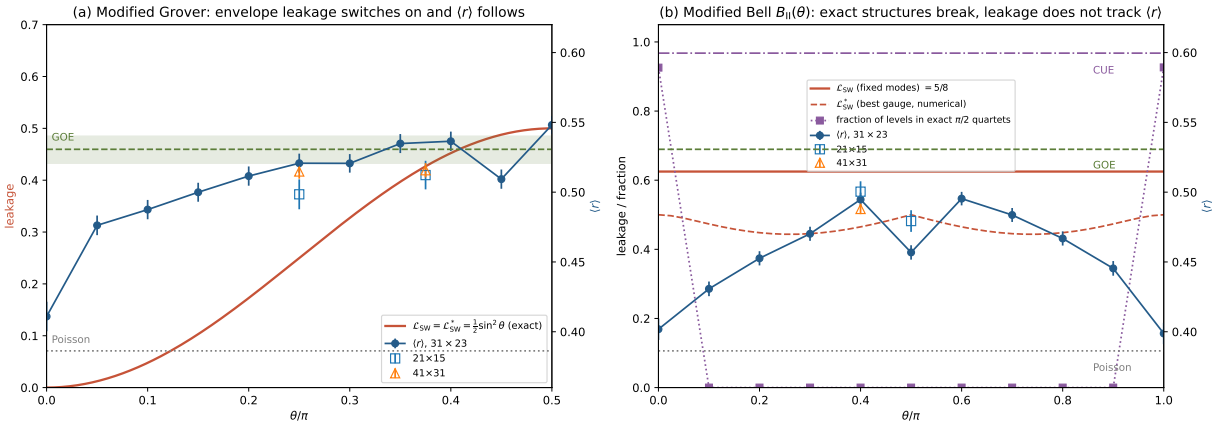


Figure 7.1: (a) Modified Grover: the exact envelope leakage $\frac{1}{2}\sin^2\theta$, which equals its gauge minimum, and the resolved $\langle r \rangle$ on 31×23, with size checks. The band is the finite-size GOE null. (b) Modified B_{II}: the fixed-mode leakage is constant at 5/8; the numerical gauge minimum stays between 0.444 and 0.5; the fraction of levels in exact $\pi/2$ quartets drops from 0.93 to zero for every interior θ; $\langle r \rangle$ stays between the Poisson and orthogonal values.

7.7 A common language, not yet a common mechanism

All the routes can be written in one language. Every demonstrated solvable family is a label algebra supplied by the geometry and the shift, and integrability is broken *within that family* when the coin makes U leak out of the algebra. What differs between the routes is which algebra is lost:

- modified Grover loses the standing-wave algebra and all its gauge images;
- the near-monomial family loses the position–direction basis;
- C_{rot} loses the row/column algebra;
- modified Bell loses an unidentified structure, visible in the spectrum as quartets and pairing.

The spectral outcomes also differ. The first two reach orthogonal-class-like local statistics at strong deformation; the last two stay intermediate at the tested sizes.

Unresolved. Whether these are different mechanisms, or one mechanism seen through different algebras, is left for future work. The question is whether there is a single structure, fixed by the rectangle and the shift, whose loss underlies every route. Candidates suggested by the existing tools are the common commutant $\mathcal{Z}_{S,C}$ and a label algebra larger than the four listed. Natural first steps, *not attempted in this revision*, are:

1. compare $\dim \mathcal{Z}_{S,C}$ along each deformation at small sizes;
2. identify the operator or algebra behind B_{II}'s $\pi/2$ quartets and corner states;
3. test whether $\langle r \rangle$ along each route tracks the escape $\mathcal{M}^{\mathrm{alg}}$ of the algebra that route destroys.

What the 16-cell map does and does not tell us

The map tells us, coin by coin and before diagonalizing anything, whether the rectangle's standing waves survive. In a gauge-adjusted form it also detects hidden standing waves, as for the compass Fourier coin. It explains precisely why the Grover rotation breaks integrability, and why the same rotation applied to B_{II} does not act through that structure. It does not detect other exact structures, such as routing, independent axes or whatever organizes B_{II}, and no leakage value sets a universal threshold for chaotic statistics.

Appendix A

Gauge-equivalent walks and the limits of fixed-mode leakage

A.1 The verified two-phase equivalence

The moving shift has an established gauge freedom. Define

$$D_{\alpha\beta} = \mathrm{diag}(e^{-i\alpha/2}, e^{i\alpha/2}, e^{-i\beta/2}, e^{i\beta/2}), \tag{A.1}$$

$$W|x, y, d\rangle = e^{i(\alpha x+\beta y)} D_{dd}|x, y, d\rangle. \tag{A.2}$$

For a homogeneous coin,

$$\boxed{WU_C W^\dagger = U_{\bar{D}C\bar{D}}.} \tag{A.3}$$

On an allowed step the position phase supplies the corresponding opposite-direction phase. At a blocked step the position is unchanged and $D_{\bar{d}\bar{d}} = \bar{D}_{dd}$, giving the same identity. Changing the sign of the parameters shows that C and DCD give unitarily equivalent full walks. This is a congruence of the coin, realized as a similarity of the walk.

If P_a are conserved by U_C, the transformed projectors $WP_aW^\dagger$ are conserved by its gauge image. A fixed choice of untransformed modes can therefore report leakage in an exactly solvable walk.

A.2 A counterexample at the same leakage as the modified Grover coin

The already verified example uses

$$D_\phi = \mathrm{diag}(e^{i\pi/4}, e^{-i\pi/4}, 1, 1), \qquad C' = D_\phi G D_\phi. \tag{A.4}$$

Its fixed-mode leakage is $3/8$, the same as $C_G(\pi/3)$, but with

$$W|x, y, d\rangle = e^{i\pi x/2}(D_\phi)^*_{dd}|x, y, d\rangle \tag{A.5}$$

one has $U_{C'} = WU_G W^\dagger$. It has exactly Grover's spectrum and an explicitly transformed complete mode family. This is an exact counterexample to a universal chaos threshold based on $\mathcal{L}_{\mathrm{SW}}$ alone.

The gauge-adjusted version is

$$\boxed{\mathcal{L}^*_{\mathrm{SW}}(C) = \min_{D=\mathrm{diag}(e^{ia},e^{-ia},e^{ib},e^{-ib})} \mathcal{L}_{\mathrm{SW}}(DCD).} \tag{A.6}$$

It tests membership in the gauge orbit of the specified standing-wave family. A numerical minimizer gives an upper bound on the true minimum. A positive value found numerically is not a proven global lower bound.

A.3 The established exact minimum for modified Grover

The supplied teaching source contains the prior exact derivation

$$\boxed{\mathcal{L}^*_{\mathrm{SW}}(C_G(\theta)) = \tfrac{1}{2}\sin^2\theta.} \tag{A.7}$$

For completeness, put $u = \sin^2 a$, $v = \sin^2 b$, and $t = \cos^2\theta$. The retained weight in Eq. (6.10) after the gauge is

$$F = 4(1 - \mathcal{L}_{\mathrm{SW}}(DCD)) = 2 + 2t - 2(1+t)(u+v) + 2t(u^2+v^2) + (4-2t)uv. \tag{A.8}$$

It is separately convex in $u, v \in [0,1]$, so a maximum occurs at a vertex. The four values are $2+2t, 2t, 2t, 2$, proving the result. This exact proof belongs to the prior project work; no new minimum search was performed for these notes.

The Hadamard-per-axis coin $H \oplus H$, with $H = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)/\sqrt{2}$, has fixed leakage $1/2$ but gauge-minimized leakage zero. The choice $a = b = \pi/4$ makes its coherent matrix diagonal. Thus independent-axis structure and gauge-transformed mode structure can overlap as solvable families.

A.4 An exactly solvable coin with half its weight in forbidden cells

The compass Fourier coin is a second, simpler example of the same point. Its fixed-mode forbidden weight is $\mathcal{L}_{\mathrm{SW}}(F_c) = 1/2$, because it exchanges v_+ with h_-. With $D = \mathrm{diag}(i, -i, 1, 1)$, however, $\widetilde{DF_cD} = (i\sigma_x) \oplus (-1) \oplus (i)$ has no forbidden weight. The gauge transformation $W|x,y,d\rangle = (-1)^x \mathrm{diag}(-i, i, 1, 1)_{dd}|x,y,d\rangle$, which is the case $\alpha = \pi$, $\beta = 0$ of Eq. (A.3), maps the walk exactly onto the transformed one (Section 6.7). This supplies the explicit transformation that the earlier record had identified only numerically.

A.5 Four demonstrated reductions, not a classification theorem

The project has explicit solvability arguments for:

1. independent homogeneous one-dimensional rows and columns when $C = C_H \oplus C_V$;
2. fixed standing-wave blocks for $\widetilde{C} = C_+ \oplus e^{i\alpha} \oplus e^{i\beta}$;
3. gauge images of the second family, for example the compass Fourier coin;
4. position–direction routing for monomial coins.

They need not be disjoint, and they have not been proved exhaustive. For independent rows and columns the block sizes grow with a side length, and the reduction is to separate one-dimensional problems; for the standing-wave family the blocks have size at most four.

Appendix B

Advanced symmetry and mode diagnostics already established

This appendix collects tools from the existing project that are useful after the first teaching sequence. They generalize the simple leakage without turning it into a universal chaos criterion.

B.1 Allowing deterministic relabeling of sectors

Equation (6.8) vanishes only if each sector keeps its original label. To accept deterministic exchanges of sectors, use

$$\boxed{\mathcal{L}^{\mathrm{route}}_{\mathscr{P}}(U) = 1 - \frac{1}{D} \max_{\pi \text{ permutation} \atop r_{\pi(a)}=r_a} \sum_a \left\| P_{\pi(a)} U P_a \right\|_F^2 .} \tag{B.1}$$

Its zero is equivalent to $UP_aU^\dagger = P_{\pi(a)}$ for an equal-rank permutation. The sum of projectors around a routing cycle is conserved. For the two equal-rank coin groups, $\mathcal{L}^{\mathrm{route}}_q = \mathcal{M}_q/2$.

The earlier restricted summary $\min(\mathcal{L}^{\mathrm{route}}_{\mathrm{SW}}, \mathcal{L}^{\mathrm{route}}_{\mathrm{basis}})$ recognized two particular reductions. Enlarging the specified family to include rows/columns and gauge images is meaningful, but a positive minimum excludes only the tested reductions.

B.2 One formula for the escape of a label algebra

For $m > 1$ complete sector projectors, define the algebra of observables constant inside each sector and its normalized basis:

$$\mathcal{A}_{\mathscr{P}} = \mathrm{span}\{P_a\}, \qquad B_a = P_a/\sqrt{r_a}, \qquad \mathbb{E}_{\mathscr{P}}(X) = \sum_a \frac{\mathrm{Tr}(P_a X)}{r_a} P_a. \tag{B.2}$$

The map $\mathbb{E}$ replaces each diagonal block by its average; it differs from retaining the entire block $\sum_a P_a X P_a$. Conjugation $X \mapsto UXU^\dagger$ compressed to the label algebra has matrix

$$\mathsf{K}_{ba} = \mathrm{Tr}(B_b U B_a U^\dagger) = \frac{W_{ba}}{\sqrt{r_b r_a}}. \tag{B.3}$$

The established unified diagnostic is

$$\boxed{\mathcal{M}^{\mathrm{alg}}_{\mathscr{P}}(U) = \frac{m - \|\mathsf{K}\|_F^2}{m-1} = \frac{1}{m-1} \sum_a \left\| (1 - \mathbb{E}_{\mathscr{P}})(U B_a U^\dagger) \right\|_F^2 .} \tag{B.4}$$

It lies in $[0,1]$. Its zero is exactly the routing condition $UP_aU^\dagger = P_{\pi(a)}$ with equal ranks. The reason is that zero escape makes conjugation an automorphism of this commutative algebra, and such an automorphism permutes its minimal projectors. No assignment optimization is needed for this test.

For the binary groups it reduces to

$$\mathcal{M}_q^{\mathrm{alg}} = 4p(1-p) = \mathcal{M}_q(2-\mathcal{M}_q). \tag{B.5}$$

For the rank-one position–direction basis it becomes

$$\mathcal{M}_{\mathrm{basis}}^{\mathrm{alg}}(U) = \frac{D - \sum_{i,j} |U_{ij}|^4}{D-1}, \tag{B.6}$$

which vanishes for precisely the monomial unitaries. For Grover, choosing the standing-wave projectors instead gives zero. Its normalization averages label observables, rather than the state probabilities used by $\mathcal{L}$; nonzero values need not agree.

This is an application of established coherence-generation and algebra-overlap constructions [5, 6, 7], not a claimed new universal detector. With $\mathcal{B} = \mathcal{A}'_{\mathscr{P}}$, it equals $m/(m-1)$ times the geometric algebra anti-correlator $G_{\mathcal{B}}(U)$ used in the earlier report.

B.3 Finding the part of a mode structure that survives

If $\mathsf{K}z = \mu z$ with $|\mu| = 1$, then

$$Q_z = \sum_a z_a B_a \quad \Longrightarrow \quad UQ_zU^\dagger = \mu Q_z. \tag{B.7}$$

Projection cannot increase the Hilbert–Schmidt norm. Equality of the full and compressed norms for this eigenvector forces the component outside the label algebra to vanish. In particular, $\mu = 1$ identifies an exactly conserved observable. A singular value one alone only gives a one-step statement, not persistent invariance of an observable.

Equivalently, for real q_a and $Q(q) = \sum_a q_a P_a$,

$$\|[U, Q(q)]\|_F^2 = \sum_{a,b} (q_a - q_b)^2 W_{ba} \tag{B.8}$$

$$= 2q^{\mathsf{T}} \left[\mathrm{diag}(r) - \frac{W + W^{\mathsf{T}}}{2} \right] q. \tag{B.9}$$

Connect labels whenever $W_{ab} + W_{ba} > 0$. Conserved q's are precisely those constant on each connected component. On the recorded 9×7 example there are 79 standing-wave sectors. Grover leaves all 79 isolated. At $C_G(\pi/3)$ there are three components: $\mathcal{K}_{00}$, the rest of the even rotation sector, and the odd rotation sector. Large invariant components do not solve their internal dynamics.

The matrix W_{ba}/r_a has a one-step probability interpretation. Its powers do not generally give the unmeasured quantum evolution of sector probabilities: the compression discards coherences that can later return.

B.4 A systematic search for common conserved observables

The shift commutant and the common commutant are

$$\mathcal{B}_S = \{X : [X, S] = 0\}, \qquad \mathcal{Z}_{S,C} = \{X : [X, S] = [X, \mathcal{C}] = 0\}. \tag{B.10}$$

In an orthonormal operator basis E_α of $\mathcal{B}_S$, the positive matrix

$$H_{\alpha\beta} = \frac{1}{4}\,\mathrm{Tr}\left([\mathcal{C}, E_\alpha]^\dagger [\mathcal{C}, E_\beta]\right) \tag{B.11}$$

has a kernel giving exactly $\mathcal{Z}_{S,C}$, because its quadratic form is $\|[\mathcal{C}, X]\|_F^2/4$. Its eigenvalues lie in $[0, 1]$. The identity is always a trivial conserved element. This includes coherences outside the standing-wave label algebra. The prior finite 4×3 check found $\dim \mathcal{B}_S = 216$ and $\dim \mathcal{Z}_{S,C_G(\pi/3)} = 6$. These numbers are not an all-size theorem. A symmetry of the product need not commute with its factors separately, so this search does not exhaust every possible full-walk symmetry.

B.5 Uniform and checkerboard subspaces

The supplied teaching source also established the following simple spaces. Define

$$|u\rangle = N^{-1/2} \sum_{x,y} |x, y\rangle, \qquad |\chi\rangle = N^{-1/2} \sum_{x,y} (-1)^{x+y} |x, y\rangle. \tag{B.12}$$

The shift is $+1$ on $|u\rangle \otimes \mathrm{span}\{h_+, v_+\}$ and -1 on $|\chi\rangle \otimes \mathrm{span}\{h_-, v_-\}$. For the second space an interior move changes the checkerboard sign, whereas a wall reversal changes the antisymmetric direction sign. Both give the same factor -1.

Whenever $[C, J] = 0$, its coherent matrix is $\widetilde{C} = C_+ \oplus C_-$. The two displayed spaces remain invariant under the full walk, with restricted matrices C_+ and $-C_-$. For modified Grover they supply eigenvalues

$$\boxed{+1, -1, e^{i\theta}, e^{-i\theta}.} \tag{B.13}$$

The checkerboard projector generally has off-diagonal terms between the terminal standing-wave sectors, explaining why a graph of the original diagonal labels does not find it. These special states still leave an extensive unresolved space.

B.6 General reversing and spectral-equivalence tests

For a corresponding pair

$$(P, \sigma) \in \{(\mathcal{J}, J), (\mathcal{P}_x\mathcal{J}, \sigma_y), (\mathcal{P}_y\mathcal{J}, \sigma_x), (\mathcal{R}\mathcal{J}, I)\}, \tag{B.14}$$

one has $PSP = S^{-1}$. If $\sigma C \sigma = C^{\mathsf{T}}$, then the verified sufficient construction is

$$A = \mathcal{C}^\dagger P K, \qquad AUA^{-1} = U^{-1}, \qquad A^2 = I. \tag{B.15}$$

Other established comparisons are

$$\mathrm{spec}\, U_C = \mathrm{spec}\, U_{JC^{\mathsf{T}}J}, \qquad U_{\bar{C}} = \overline{U_C}. \tag{B.16}$$

A scalar coin phase rotates the quasienergies uniformly. These are useful comparisons of walks; they should not be confused with a complete list of their commuting symmetries.

B.7 Why small leakage alone does not set a crossover scale

The prior perturbative discussion used $U_G(\theta) = U_G(0)e^{-i\theta(I\otimes V)}$. For nondegenerate Grover eigenstates a, b, a first-order hybridization indicator was

$$\kappa_{ab}(\theta) = \frac{|\theta\langle a|I \otimes V|b\rangle|}{2|\sin[(\varphi_a - \varphi_b)/2]|}. \tag{B.17}$$

The numerator must be compared to a symmetry-allowed phase separation, not only to one. Grover's extensive degenerate subspaces require degenerate perturbation theory instead. The retained mode-overlap selection rule can connect many labels, with an approximate overlap $\langle f_p, g_m \rangle \simeq (4/\pi)m/(m^2 - p^2)$ in the regime tested in the export. It is not an exact finite-size identity.

Thus loss of exact mode conservation at any nonzero allowed coupling and the appearance of a random-matrix-like spectral window are different statements. No universal crossover scale was established.

Nor can unrestricted optimization over all mode bases repair this: every finite unitary has rank-one eigenprojectors with zero leakage. The tested family must be constructed from the geometry or another independently specified structure. This is why the common algebra formula still needs its family of labels.

B.8 The wall-reflection pre-test and its limited scope

The retained bulk equation is $U(k_x, k_y)u = \lambda u$. At fixed (k_y, λ), $z \det[\lambda I - U(k_x, k_y)]$, $z = e^{ik_x}$, is a polynomial of degree at most two. The determinant may vanish identically on a flat band, roots may coincide, and a root need not be a propagating real wavevector. The regular two-wave interpretation requires these exceptional cases to be excluded.

For a superposition of bulk waves the actual right-wall equation is

$$\sum_j A_j e^{ik_j N_x} \left[(Cu_j)_R - e^{ik_j} (Cu_j)_L \right] = 0. \tag{B.18}$$

The top wall replaces (R, L, k, N_x) with (U, D, q, N_y). At a regular band point these equations define partner waves and reflection amplitudes. If horizontal and vertical reflection paths close on the same four-wave set, their consistency requires

$$K_{\text{corner}} = \frac{r_x(q) r_y(k')}{r_y(k) r_x(q')} = 1. \tag{B.19}$$

When closure holds, eigenvector phase and wall-position factors cancel from the ratio. The retained tests find closure and unit corner phase for Grover and the tested gauge images; modified Grover at generic interior parameters fails closure. At $\pi/2$ it closes but has corner phase -1. The real-generator variant below closes at the tested points but has a nonunit corner phase.

This tests a particular four-wave ansatz. It is not a complete integrability criterion: routing-solvable permutations can fail it, and independent row/column walks can make it undefined. An orbit not closing within 800 tested reflections is not a proof of an infinite orbit. The proposed ray interpretation was untested and is not used to classify the walks here.

Appendix C

Supplementary verified examples and numerical records

The following material is retained for a second reading. It records already completed work, not additional computations for these notes. Finite numerical observations keep their original limitations.

C.1 A position-only description of the Grover spectrum

The Grover-billiard export also established a smaller real symmetric matrix. Let $A|\mathbf{r}\rangle = |\mathbf{r}\rangle \otimes |s\rangle$, where $|s\rangle = (1,1,1,1)^{\mathsf{T}}/2$, and let S_{ff} have the older flip-flop/self-loop convention. Then

$$T = A^{\dagger} S_{\mathrm{ff}} A = \frac{\mathrm{Adj} + \mathrm{diag}(\text{missing-neighbor counts})}{4}. \tag{C.1}$$

This is a matrix of nearest-neighbor averages with missing links replaced by staying at the site. It is an auxiliary spectral matrix, not the walker Floquet operator. For the rectangle,

$$T = \frac{1}{2}(T_x \otimes I_y + I_x \otimes T_y), \qquad \mu_{mn} = \frac{1}{2}\left(\cos\frac{\pi m}{N_x} + \cos\frac{\pi n}{N_y}\right). \tag{C.2}$$

Its eigenvectors are products of the cosine profiles. For the moving Grover walk, the dispersive phases satisfy $\cos\varphi_{mn} = -\mu_{mn}$, agreeing with Eq. (6.21); the extensive flat subspaces must still be added. For $Tf = \mu f$, $-1 < \mu < 1$, put $\lambda = \mu + i\sqrt{1-\mu^2}$ and $\mathcal{Q} = I \otimes (J\Gamma)$. The recorded full-state construction is

$$\phi = \mathcal{Q}(Af - \lambda S_{\mathrm{ff}} Af), \qquad U_G \phi = -\lambda \phi. \tag{C.3}$$

It is written before normalization; the partner branch is obtained from the conjugate eigenvalue. The export verified this construction with residual at most 7.6×10^{-16} on 7×5. For a nonsquare rectangle its parity labels agree with those of the scalar reduction. On a square, the recorded moving-Grover D_4 correspondence exchanges $A_1 \leftrightarrow B_1$ and $A_2 \leftrightarrow B_2$ between the scalar and Floquet descriptions, while retaining E. These labels must therefore be assigned using the physical full-walk symmetry projectors.

This reduction and the explicit four-dimensional mode blocks are two consistent descriptions of the same rectangular Grover solution. They should not be extended to arbitrary coins by analogy.

C.1.1 The special verified correspondence between the two shifts

The Grover-billiard export gives a stronger statement for G. With $\mathcal{Q} = I_{\text{pos}} \otimes (J\Gamma)$,

$$\boxed{U_{\text{mov},G} = -\mathcal{Q} U_{\text{ff},G} \mathcal{Q}^{\dagger}.} \tag{C.4}$$

Thus the dispersive phases map by $\varphi_{\text{mov}} = \pi - \varphi_{\text{ff}}$ on the positive branch, and the ± 1 multiplicities exchange. For the rectangle the moving-shift multiplicities are those in Chapter 6; the flip-flop ones are the reverse. This is a verified Grover-specific correspondence, not a license to transfer arbitrary fixed-coin numerical results between shifts.

C.2 Square controls and representation dimensions

The square is a special rectangle. For Grover it has the full D_4 geometry, whose irreducible representation dimensions are $1, 1, 1, 1, 2$, commonly labeled A_1, A_2, B_1, B_2, E. A two-dimensional irrep can occur many times. Its isotypic subspace contains a repeated symmetry copy of each associated eigenvalue; one row of the irrep is used for an independent spacing sequence. For example, in the recorded 60×60 Grover calculation the dispersive E row contains 900 levels. Calling E a two-dimensional representation does not say the full sector contains only two states.

Modified Grover generally retains a smaller square group generated by $\mathcal{R}$ and diagonal reflection Σ: $\{I, \mathcal{R}, \Sigma, \mathcal{R}\Sigma\}$. On 23×23 its four joint sectors have dimension 529. The source's control is

Parameter	Only rotation resolved	Four joint sectors resolved
$\theta = \pi/3$	0.3644, 0.3741	0.5330, 0.5266, 0.5371, 0.5149
$\theta = \pi/2$	0.4191, 0.4213	0.5288, 0.5288, 0.5295, 0.5283

These are different symmetry resolutions, not contradictory spectra. Independent random superposition models are only comparison references for symmetry sectors that may be correlated in the deterministic walk.

C.3 The real-generator variant and the independent-axis family

The real variant of modified Grover has coherent matrix

$$\widetilde{C}'(\theta) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \begin{pmatrix} -\cos\theta & \sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix}. \tag{C.5}$$

It is real but generally nonsymmetric. It has the same fixed-mode leakage $\frac{1}{2}\sin^2\theta$, survives rotation, and satisfies $\sigma_y C' \sigma_y = C'^{\mathsf{T}}$, giving time reversal via $\mathcal{P}_x \mathcal{J}$. The recorded half-circle means on 25×19 at $\pi/3$ are $0.525, 0.536$. The earlier prediction of a unitary-class benchmark from its nonsymmetry was retracted after this reversing relation was found.

For another deterministic construction, set

$$R_\eta = \begin{pmatrix} \cos\eta & -\sin\eta \\ \sin\eta & \cos\eta \end{pmatrix}, \qquad B_\eta = R_\eta \oplus R_\eta, \qquad \eta = \pi/4, \tag{C.6}$$

and use the permutation P from Chapter 5. The recorded family is

$$C_{\text{rot}}(\varphi) = B_\eta e^{-i\varphi P}. \tag{C.7}$$

At zero it preserves independent rows and columns. It leaves that specified decomposition with exact leakage $\mathcal{L}_{\mathrm{RC}} = \sin^2 \varphi$. Its pair vector is

$$(2 \min(\cos^2 \varphi, \sin^2 \varphi), 1, 1). \tag{C.8}$$

The starting coin has $\mathcal{L}^*_{\mathrm{SW}} = 1/2$, so it supplies a genuinely different tested reduction from gauge-transformed standing waves. At general parameter the ordinary rotation does not survive, but $JCJ = C^{\mathsf{T}}$ gives time reversal and $\Gamma \bar{C} \Gamma = C$ gives a commuting antiunitary. An additional anticommuting operator was found in the source search. No complete symmetry classification was obtained.

The recorded comparison uses half circles, with rotation sectors resolved for modified Grover:

	25×19		29×18		40×13	
Parameter	C_G	C_{rot}	C_G	C_{rot}	C_G	C_{rot}
0.05	0.443	0.441	0.465	0.443	0.485	0.448
0.10	0.474	0.446	0.473	0.443	0.508	0.488
0.30	0.502	0.446	0.511	0.436	0.536	0.474
0.60	0.519	0.428	0.521	0.413	0.528	0.441
1.25	0.511	0.393	0.535	0.370	0.522	0.391

Over the source's 12 points in $[0.6, 1.3]$, its recorded aggregate means were 0.5261 for modified Grover and 0.4160 for the rotation family. The latter does not prove integrability or imply that an unidentified conserved quantity must exist.

C.4 Other explicitly tested deterministic constructions

To preserve the candidate-search record without replacing the main examples, the following definitions are retained. The Fourier-based constructions of the earlier record (fractional Fourier powers and the designed Fourier family) are removed from these notes. Put $E(a, b) = |a\rangle\langle b| + |b\rangle\langle a|$, $Y = E(R, D) + E(L, U)$, and the signed turn $X_F = E(R, U) - E(L, D)$, with $X_F^2 = I$. For two-dimensional blocks let

$$K(\eta, \xi) = e^{i\eta\sigma_x/2} R_\xi e^{i\eta\sigma_x/2}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{C.9}$$

in this formula only. It is a block Pauli matrix, not the full four-direction reflection symbol used elsewhere.

Source label	Retained definition
$C_x(\varphi)$	$e^{-i\varphi Y/2} B_{\pi/4} e^{-i\varphi Y/2}$
$C_n(\varphi)$	$e^{-i\varphi Y/2} [K(\pi/3, \pi/4) \oplus K(\pi/3, \pi/4)] e^{-i\varphi Y/2}$
$C_m(\varphi)$	Same sandwich, with blocks $K(\pi/3, \pi/4)$ and $K(\pi/5, 0.9)$
Direction hopping A	$\exp[-i(P + 0.6Y + 0.8\Gamma)]$
Direction hopping B	$\exp[-i(P + 0.6Y + 0.8\Gamma + 0.5X_F)]$
Near-monomial	$e^{-i\epsilon H/2} P e^{-i\epsilon H/2}$, $H = X_F + 0.7Y + 0.4\Gamma + 0.3E(R, L)$

Selected recorded observations are:

Family	Setting	Recorded $\langle r \rangle$ and scope
C_x	$\varphi = 0.4, 0.8, 1.2$	0.418, 0.443, 0.485 on 25×19 (session-only)
C_n	Same parameters	0.407, 0.439, 0.446 on 25×19 (session-only)
C_m	Same parameters	0.466, 0.469, 0.431 on 25×19 (session-only)
Direction hopping A	25×19, rotation sectors	0.499, 0.496
Direction hopping B	25×19	0.500
Near-monomial	$\epsilon = 0.02, 0.2, 0.8$	0.18, 0.35, 0.517 on 25×19

These are finite candidate tests, not a classification or a universal threshold. Exact monomial coins were verified to have one nonzero entry per full-walk column, consistent with the exact routing proof.

C.5 Aspect ratios, special phases, and distribution tests

The modified-Grover aspect sweep used 33 parameter values and the five rectangles in the next table. Its plateau summary averages $\theta \in \{0.6, 0.8, \pi/3, 1.25\}$:

Rectangle	Aspect ratio	Plateau mean	SD across four parameters
25×19	1.32	0.5140	0.0032
29×18	1.61	0.5354	0.0110
33×16	2.06	0.5312	0.0087
40×13	3.08	0.5287	0.0050
51×10	5.10	0.5495	0.0112

Across these 20 points the recorded average is 0.5318, with SD 0.0139. Geometry-dependent offsets of order 0.01–0.02 were not excluded. At $\pi/3$, the source's pooled distribution has $n = 4963$, mean 0.5360, and a calibrated COE test $p = 0.080$; the uncalibrated independent-sample test gave $p = 0.009$. Its full archive was not attached to the earlier audit review described in the supplied notes. These are retained as recorded results of the exported session, not claimed as a fresh independent reproduction.

The suggested $0.35/\sqrt{n}$ scale was calibrated on finite COE simulations. It is not a universal error formula. Taking one conjugate half removes exact repetitions but does not make neighboring ratios independent. Full- and half-circle means can differ through endpoint gaps. No confidence interval is inferred here from the sweep SDs.

The source observed additional degenerate phases away from Grover. For example, the full-spectrum multiplicity of each of ± 1 at $\theta = \pi/3$ was 1 on 25×19, 13 on 40×13, 29 on 29×18, and 42 on 26×18. The general side-parity pattern was empirical, not proved. At $\pi/2$, $\pm i$ also show parity-dependent degeneracies. A source table of these special-point means is summarized by the odd-by-odd versus even-side distinction in the main chapter.

C.6 Random coins, flux, and frozen-disorder controls

The additional Haar-coin study reported four means on 25×19: $0.556, 0.555, 0.520, 0.486$, pooled mean 0.5291. The coin generation was complex Gaussian QR with phase correction. Restricted searches found no time reversal, yet the local ratios were closer to the orthogonal reference than the unitary reference. This is an unexplained observation; the searches are not absence proofs.

A flux-control step was specified as

$$U_B = S\ \mathrm{diag}_{x,y,d}(\mathrm{ph})\,[I \otimes C_G(\pi/3)], \qquad \mathrm{ph} = (e^{iB(y-c)}, e^{-iB(y-c)}, 1, 1), \tag{C.10}$$

with $c = (N_y - 1)/2$ in the zero-based implementation. On 25×19, the recorded sector means were $(0.527, 0.500)$ at $B = 0$, $(0.551, 0.579)$ at $B = 0.05$, and $(0.552, 0.536)$ at $B = 0.2$. Here B is the phase parameter as written in the source code, without an added conversion of units. These samples did not demonstrate unitary-class universality.

Frozen, site-dependent rotation-family coins were also tested on 25×19, changing the homogeneous model. Random $\varphi_{xy} \in [0.3, 0.9]$ gave 0.533; random $\eta_{xy} \in \pi/4 \pm 0.3$ gave 0.516, compared with 0.428 in the homogeneous reference. These are finite-size controls. They do not establish a thermodynamic delocalized phase, and the exported discussion explicitly qualified the original "disorder restores chaos" interpretation.

Appendix D

The older random-coin interpolation uses a different shift

This appendix consolidates the two earlier crossover exports. Their results are useful, but they belong to the flip-flop/self-loop shift of Chapter 2. They are not moving-shift results for the same unchanged coin.

D.1 A second shift used in the older sessions

Some exports instead use a flip-flop shift S_{ff}. It reverses the direction after an allowed move and keeps the state unchanged at a missing link. The two conventions are:

	Moving shift S_Ω	Flip-flop shift S_{ff}
Allowed link	$\lvert\mathbf{r}, d\rangle \mapsto \lvert\mathbf{r} + \delta_d, d\rangle$	$\lvert\mathbf{r}, d\rangle \mapsto \lvert\mathbf{r} + \delta_d, \bar{d}\rangle$
Missing link	$\lvert\mathbf{r}, d\rangle \mapsto \lvert\mathbf{r}, \bar{d}\rangle$	$\lvert\mathbf{r}, d\rangle \mapsto \lvert\mathbf{r}, d\rangle$
Square	Generally $S_\Omega^2 \neq I$	$S_{\mathrm{ff}}^2 = I$

The verified relation is

$$S_\Omega = \mathcal{J} S_{\mathrm{ff}}. \tag{D.1}$$

This does not mean that an arbitrary fixed coin gives the same full walk under both shifts. The old random-coin interpolation is therefore kept separate in Appendix D. A special verified spectral correspondence for Grover is given in Appendix C.

The historical shift convention is a different model

"Reflecting boundary" is insufficient to reconstruct a walk. The internal direction after an allowed move and after a blocked move must both be stated. The involution property of the flip-flop shift must not be assigned to the moving shift.

D.2 The specified historical model

The geometry is $N_x = 10, N_y = 6$, $D = 240$, with no periodic boundary. The step is

$$U_{\mathrm{ff}}(\lambda) = S_{\mathrm{ff}}[I_{60} \otimes C(\lambda)]. \tag{D.2}$$

The shift satisfies $S_{\mathrm{ff}}^\dagger = S_{\mathrm{ff}}$, $S_{\mathrm{ff}}^2 = I$, and $\mathrm{Tr}\, S_{\mathrm{ff}} = 2N_x + 2N_y = 32$, the boundary self-loops. The original coin was a single COE draw $V^{\mathsf{T}}V$, with V obtained from a seeded complex-Gaussian QR

construction, seed 16. E3 records that seed 16 was selected after scanning seeds 1–30 for proximity to the orthogonal reference. Their recorded mean was 0.4884935, population SD 0.0371222, minimum 0.4132964 and maximum 0.5499779. The selected 0.5318896 is not an ensemble mean. The preliminary seed-20261001 endpoint was 0.4181041 with the flip-flop shift and 0.4261382 in its exploratory moving-shift test. Those are distinct records, not points on the seed-16 curve.

The supplied reproduction instead specifies the following six-decimal real symmetric generator as its numerical input:

$$K_*^{\mathrm{doc}} = \begin{pmatrix} 1.154349 & -0.635057 & 0.212309 & -0.148986 \\ -0.635057 & 0.875129 & 0.203120 & -1.258544 \\ 0.212309 & 0.203120 & 1.490150 & 1.547436 \\ -0.148986 & -1.258544 & 1.547436 & -0.761511 \end{pmatrix}. \tag{D.3}$$

The split is

$$K(\lambda) = \begin{pmatrix} K_H & \lambda W \\ \lambda W^{\mathsf{T}} & K_V \end{pmatrix}, \qquad C(\lambda) = e^{-iK(\lambda)}, \qquad 0 \le \lambda \le 1, \tag{D.4}$$

where K_H, K_V, W are the displayed generator's blocks. It couples the two axis sectors of the *generator*. It is different from the path $e^{-i\lambda K_*}$, which scales the whole generator.

The coins are symmetric unitary, and the reproduced full step has time reversal $A_{\mathrm{ff}} = S_{\mathrm{ff}}K$, with square one. At zero the coin is block diagonal in the axes. The row and column projectors commute with the full step. Its spectrum contains 20 distinct phases repeated six times and 12 distinct phases repeated ten times in this example, as checked in the reproduction.

D.3 The degenerate endpoint and the definition of a ratio mean

The full zero-coupling spectrum has 208 zero gaps. There are 176 undefined 0/0 ratios and 64 ratios equal to zero next to a nonzero gap. The two exports use different estimators at this point:

Convention	Treatment of the $\lambda = 0$ full spectrum
Strict formula	The mean including every ratio is undefined
Omit undefined ratios	The retained mean is zero (numerical original: about 7.7×10^{-14})
Replace undefined ratios by zero	The zero-filled mean is zero, but this is a different estimator

None is evidence of a Poisson endpoint. Removing 0/0 terms is not symmetry resolution. The degeneracy comes from identical homogeneous row and column copies.

D.4 Recorded positive-coupling values and precision labels

The original full-precision seeded construction and the reproduction's six-decimal generator are different specified numerical inputs. Their small differences are retained rather than silently replacing one data record by the other:

λ	Original seeded path	Six-decimal documented path
0.02	0.297681	0.297632
0.10	0.361189	0.361190
0.12	0.393100	0.393099
0.20	0.445458	0.445458
0.40	0.483130	0.483131
0.60	0.483969	0.483968
0.80	0.511780	0.511779
1.00	0.531890	0.531890

No symmetry sectors were resolved in these historical full-circle values. The reproduction checked the documented path in independent Python and Octave implementations, with agreement about 3×10^{-12} at a representative checkpoint and 56 Octave assertions passed. The original positive-coupling data are redrawn below from their retained 51-point CSV. No new spectra are involved.

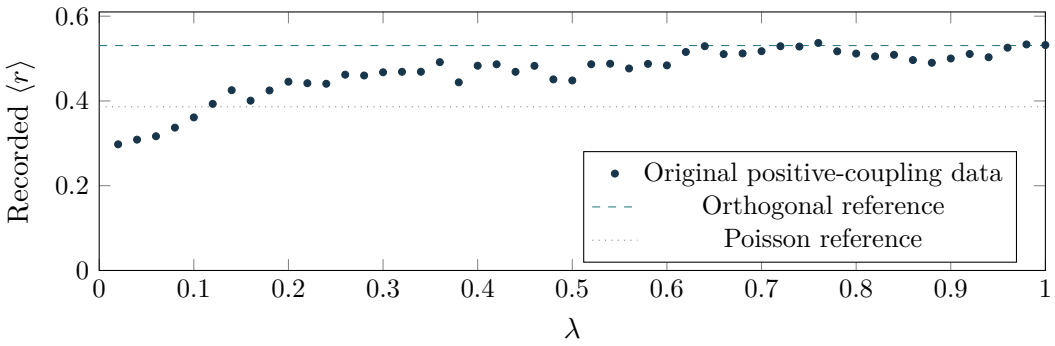


Figure D.1: The historical 10×6 flip-flop/self-loop walk, redrawn from the original exported CSV. Only positive coupling is shown; its degenerate zero-coupling spectrum needs the estimator qualification in the text. This is not the main moving-shift model.

These data demonstrate a degenerate decomposable endpoint and orthogonal-class-like local means for one finite realization, not a universal Poisson-to-COE crossover. The disputed alternative plot was not reproduced by this model, a whole-generator scaling, or periodic boundaries. A consistent change of site-index order cannot change eigenphases. The special correspondence for Grover between the two shifts is given in Appendix C.

Appendix E

Source inventory, consistency decisions, and remaining limits

E.1 Sources and their distinct roles

The supplied set contains five distinct Markdown exports, one teaching source and two distinct PDFs. Three pairs of export files and the two teaching-PDF copies are byte-identical. Duplicate attachments are not independent replications.

Source	Attached filename	Role
E1	`DQW_chat_complete_export(1).md`	Original 10×6 flip-flop interpolation, complete CSV and estimator
E2	`DQW_session_export(1).md`	Rectangle audit: coins, exact modes, diagnostics, modified Grover, designed Fourier and numerical qualifications
E3	`DQW_this_chat_self_contained_export(1).md`	Earlier interpolation including the 30-seed scan, seed selection and preliminary moving-shift run
E4	`DQW_chat_export(1).md`	Both Grover conventions, rectangular spectral mapping, direct sector checks and square controls
E5	`DQW_chat_export_2026-10-17(1).md`	Independent documented-path reproduction, estimator comparison and disputed-plot checks
E6	Computations for this revision (no separate file)	Coin-level 16-cell analysis, Fourier gauge, two Bell matrices, IPR reproductions, 31×23 Grover and Bell sweeps (computed for this revision)
T	`DQW_rectangle_teaching_draft(1).tex`	Existing teaching source with advanced label algebra, common commutant, checkerboard and exact gauge-minimum derivations
R	`DQW_rectangle_revised(1).pdf`	Revised audit report, qualified using E2's verification labels

E1 and E3 overlap in their original work, rather than supplying independent replication. E5

reports its Python/Octave reproduction. E2 distinguishes its reproduced core results from later session-only logs; these notes preserve that distinction. Additional identities from T are retained with their displayed prior derivations. Its references to other write-ups do not mean unprovided files were independently opened for this update.

E.2 Figures and their origin

All figures are in the folder `Figures`. Figures 4.1–5.2 are exact coin-level drawings; the others were computed for this revision (Python/NumPy, moving shift, direction order as in the text).

File	Content
`fig01_wall_envelopes_mask.pdf`	Wall reversal, envelopes, allowed/forbidden cells
`fig02_reading_the_square_grover.pdf`	Physical and coherent squares of Grover
`fig03_base_coins_16cells.pdf`	All unperturbed coins in both squares
`fig04_deformed_coins_16cells.pdf`	Modified Grover and modified Bell squares
`fig05_mechanisms_leakage_vs_r.pdf`	Leakage, gauge minimum, quartets and $\langle r \rangle$ along both deformations
`fig06_ipr_bellII_grover_40x35.png`	IPR reproductions, 40×35
`fig07_grover_sectors_40x35.png`	Grover sector-resolved IPR, staircase, DOS
`fig08_bellII_spectrum_40x35.png`	B_{II} IPR, staircase, DOS
`fig09_ipr_collage_modified_grover.png`	IPR collage of the modified Grover walk
`fig10_ipr_collage_modified_bellII.png`	IPR collage of the modified Bell walk
`fig11_r_vs_theta.png`	$\langle r \rangle$ versus θ for both families

E.3 Topic map and verification status

Topic	Sources	Status retained here
Space, coin and ordered Floquet product	E1–E5, T, R	Definitions and algebraic unitarity
Moving shift and direction reverser	E2, T, R	Exact identities and reported checks
Rectangle cycles, period, direct sums	E2, T, R	Explicit construction and audit
Combined reflections	E2, E4, T	Operators and recorded commutators
Standing waves, terminal modes, projectors	E2, T, R	Exact formulas and reported core audit
Splitting, pair mixing, axis entropy	E2, T, R	Distinct definitions and verified relations
Coins and Fourier assignments	E2, T, R, E6	Explicit matrices; two Bell matrices distinguished
16-cell map and envelope rule	T, E2, E6	Exact; new explanatory figures
Permutation/monomial routing	E2, T, R	Exact cycle solution
Grover blocks, phases, flat counts	E2, E4, T	Exact solution and direct checks
Bell boundary and mode leakage	E2, T, R	Verified diagnostics; full classification open
Modified-Grover leakage and symmetries	E2, T, R	Exact identities and numerical checks

Topic	Sources	Status retained here
Modified-Grover original ratio table	E2, T, R	Independent reproduction reported
Aspect, onset and distribution tests	E2, T, R	Finite evidence; session-only subsets labeled
Special-point parity anomaly	E2, T, R	Displayed symmetries; mechanism unresolved
Compass Fourier gauge solution	E2, E6	Explicit gauge and block check (computed for this revision)
Bell B_{II} and modified Bell	E6	Computed for this revision; classification open
IPR reproductions and 31×23 sweeps	E6	Computed for this revision; filters stated
Gauge and transpose equivalences	E2, T, R	Prior proofs and recorded checks
Modified-Grover exact gauge minimum	T	Existing derivation reproduced
Algebra, common commutant, simple spaces	T	Existing constructions; finite checks stay size-specific
Wall-reflection pre-test	E2, T, R	Restricted ansatz; no completeness theorem
Other coins, Haar, flux, disorder	E2, T, R	Supplementary finite records, qualified interpretations
Historical COE interpolation	E1, E3, E5	Different shift; selected endpoint not universal evidence

No extra research source was retrieved. The only new calculations are the session results of E6, labeled as such in the text. E4's disk and triangle data are outside the rectangle-only scope. Its square results are explicitly labeled controls.

E.4 Decisions needed to make a consistent teaching text

1. **One physical direction order.** Every displayed main coin is in (R, L, U, D). The (U, D, R, L) Grover export is translated at the level of named directions. The two Fourier assignments are kept distinct because the shift is held fixed.
2. **One main shift.** The moving, stay-and-reverse shift is used throughout the core. The old flip-flop/self-loop data retain their own convention and are not relabeled as main-model data.
3. **One step order.** Position space comes first in the tensor product, and $U = S(I \otimes C)$. Alternative site-index orders are basis permutations if applied consistently.
4. **Exact structure versus numerical interpretation.** Grover and monomial walks have displayed exact reductions. Bell does not have a demonstrated complete reduction in the verified record. Its integrability is not assumed.
5. **Three meanings of mixing.** Individual-direction splitting, a coarse pair label, and a selected spatial-mode label are distinguished. A zero of the minimum pair score is not identified with a monomial coin.
6. **No universal scalar test.** Mode leakage is always tied to its projectors, and gauge changes can move a conserved family. A positive value excludes that particular preservation condition.

7. **Degeneracy conventions stay visible.** Undefined ratios, discarded flat phases, selected branches, and unresolved sectors are not silently converted into evidence of Poisson or COE statistics.
8. **Gauge identifications are explicit where available.** The compass Fourier coin now has an explicit gauge transformation onto the standing-wave family (Section 6.7); the modified-Grover minimum has the exact proof of Appendix A. Other gauge minima quoted for $P, F_p, B_{\mathrm{I}}, B_{\mathrm{II}}$ are numerical upper bounds.
9. **Two Bell matrices.** B_{I} (earlier report) and B_{II} (QWProgramming Bell basis, reproduces the IPR figure) are different coins, not related by any signed relabeling. Results are never transferred between them.
10. **The designed Fourier family is removed** from the notes at the project's request; only the unperturbed Fourier coins are kept.

E.5 What the consolidated record does not establish

The supplied material does not classify every integrable coin. It does not settle Bell's complete dynamics or the mechanism of the modified-Grover parity anomaly. A finite failed symmetry search does not exclude a nonlocal integral. The wall-reflection pre-test does not prove completeness of four-wave eigenstates.

The rectangle studies of the modified coins establish nearest-neighbor spectral evidence over the reported finite sizes and windows. They do not supply an all-size theorem, a universal threshold in either mixing measure, or a complete spectral-form-factor and eigenvector characterization. The long-range studies of other geometries in E4 do not fill that gap for the rectangle.

The older selected-coin interpolation is not an ensemble-averaged curve or a proof of a universal crossover. E3's 30-seed endpoint scan is quoted separately; it is not an ensemble curve at every coupling. Proposed integrability-preserving phase offsets were not verified in those exports and are not adopted as established examples. No novelty claim is made in these teaching notes, and no new literature search was performed for it.

References

[1] Project session exports E1–E5 listed in Appendix E. These are the immediate sources of the numerical records in this consolidation. They are not five independent replications.

[2] Y. Y. Atas, E. Bogomolny, O. Giraud, and G. Roux, *Distribution of the Ratio of Consecutive Level Spacings in Random Matrix Ensembles*, Physical Review Letters **110**, 084101 (2013). https://doi.org/10.1103/PhysRevLett.110.084101.

[3] Y. Higuchi, E. Segawa, and A. Suzuki, *Spectral mapping theorem of an abstract quantum walk*, https://arxiv.org/abs/1506.06457. Referenced in E4 for the auxiliary Grover spectral reduction.

[4] N. Konno, I. Sato, E. Segawa, and Y. Shikano, *Quantum walks driven by quantum coins with two multiple eigenvalues*, Quantum Studies: Mathematics and Foundations **10**, 41–65 (2023). https://doi.org/10.1007/s40509-022-00281-1. Its bibliographic details were confirmed in the earlier discussion, after E2 had left them unverified.

[5] P. Zanardi, G. Styliaris, and L. Campos Venuti, *Coherence generating power of quantum unitary maps and beyond*, Physical Review A **95**, 052306 (2017). https://doi.org/10.1103/PhysRevA.95.052306.

[6] P. Zanardi and L. Campos Venuti, *Quantum coherence generating power, maximally abelian subalgebras, and Grassmannian geometry*, Journal of Mathematical Physics **59**, 012203 (2018). https://doi.org/10.1063/1.4997146.

[7] P. Zanardi, *Quantum scrambling of observable algebras*, Quantum **6**, 666 (2022). https://doi.org/10.22331/q-2022-03-11-666.